



QIIME2-ITS run summary

Samples: 53

Default report grouping column: country

(alpha/beta diversity figures below also cover any other column with a statistically significant result)

Median non-chimeric reads/sample: 93386

Run information

Run started:	2026-09-21T12:41:22.505694-04:00
Run finished:	2026-09-21T16:09:20.685999-04:00
Run duration:	3h27m58s
Run by:	(not recorded)
Platform:	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Conda environment:	rachis-qiime2-2026.7
qiime2-its version:	0.3.2
QIIME2 framework version:	2026.7.0
BBMap (bbduk.sh) version:	39.80
Python version:	3.12.13
Input folder:	raw_reads
Metadata file:	metadata.tsv
Classifier file:	classifier/unite-classifier.qza
Output folder:	output
Command invoked:	<pre>qiime2-its \ -q rachis-qiime2-2026.7 \ -i raw_reads \ -o output \ -m metadata.tsv \ -c classifier/unite-classifier.qza \ -pe \ --extract-its2 \ --taxa Fungi \ -t 40 \ -p 10</pre>

Pipeline parameters

parameter	value
allow-one-off	False
chimera-method	consensus
extract-its1	False
extract-its2	True
max-ee	2.0
max-ee-r	None
max-len	0
max-rarefaction-depth	4000
min-fold-parent-over-abundance	1.0
min-len	0
n-reads-learn	1000000
parallel-processes	10
pe	True
pooling-method	independent
report-metadata-column	None
reverse-complement	False
sampling-depth	1000
se	False
skip-advanced-stats	False
skip-report	False
taxa	Fungi
threads	40
trunc-q	2

Input sample files

sample	file
K.BeL.1.1	K.BeL.1.1_S1_L001_R1_001.fastq.gz
K.BeL.1.1	K.BeL.1.1_S1_L001_R2_001.fastq.gz
K.BeL.1.2	K.BeL.1.2_S2_L001_R1_001.fastq.gz
K.BeL.1.2	K.BeL.1.2_S2_L001_R2_001.fastq.gz
K.CSL.1.1	K.CSL.1.1_S3_L001_R1_001.fastq.gz
K.CSL.1.1	K.CSL.1.1_S3_L001_R2_001.fastq.gz
K.CSL.1.2	K.CSL.1.2_S4_L001_R1_001.fastq.gz
K.CSL.1.2	K.CSL.1.2_S4_L001_R2_001.fastq.gz
K.CT9.1.1	K.CT9.1.1_S5_L001_R1_001.fastq.gz
K.CT9.1.1	K.CT9.1.1_S5_L001_R2_001.fastq.gz
K.CtE.1.2	K.CtE.1.2_S6_L001_R1_001.fastq.gz
K.CtE.1.2	K.CtE.1.2_S6_L001_R2_001.fastq.gz
K.Ira.13.2	K.Ira.13.2_S7_L001_R1_001.fastq.gz
K.Ira.13.2	K.Ira.13.2_S7_L001_R2_001.fastq.gz
K.KKF.1.1	K.KKF.1.1_S8_L001_R1_001.fastq.gz
K.KKF.1.1	K.KKF.1.1_S8_L001_R2_001.fastq.gz
K.KKF.1.2	K.KKF.1.2_S9_L001_R1_001.fastq.gz
K.KKF.1.2	K.KKF.1.2_S9_L001_R2_001.fastq.gz
K.KKF.2.1	K.KKF.2.1_S10_L001_R1_001.fastq.gz
K.KKF.2.1	K.KKF.2.1_S10_L001_R2_001.fastq.gz
K.KKF.2.2	K.KKF.2.2_S11_L001_R1_001.fastq.gz
K.KKF.2.2	K.KKF.2.2_S11_L001_R2_001.fastq.gz
K.Leg.1.1	K.Leg.1.1_S12_L001_R1_001.fastq.gz
K.Leg.1.1	K.Leg.1.1_S12_L001_R2_001.fastq.gz
K.Leg.1.2	K.Leg.1.2_S13_L001_R1_001.fastq.gz
K.Leg.1.2	K.Leg.1.2_S13_L001_R2_001.fastq.gz
K.WKJ.2.1	K.WKJ.2.1_S14_L001_R1_001.fastq.gz
K.WKJ.2.1	K.WKJ.2.1_S14_L001_R2_001.fastq.gz
K.WKJ.2.2	K.WKJ.2.2_S15_L001_R1_001.fastq.gz
K.WKJ.2.2	K.WKJ.2.2_S15_L001_R2_001.fastq.gz
K.ZER.1.1	K.ZER.1.1_S16_L001_R1_001.fastq.gz
K.ZER.1.1	K.ZER.1.1_S16_L001_R2_001.fastq.gz
K.ZER.1.2	K.ZER.1.2_S17_L001_R1_001.fastq.gz
K.ZER.1.2	K.ZER.1.2_S17_L001_R2_001.fastq.gz
K.ZER.2.1	K.ZER.2.1_S18_L001_R1_001.fastq.gz
K.ZER.2.1	K.ZER.2.1_S18_L001_R2_001.fastq.gz
K.ZER.2.2	K.ZER.2.2_S19_L001_R1_001.fastq.gz
K.ZER.2.2	K.ZER.2.2_S19_L001_R2_001.fastq.gz
K.ZST.1.1	K.ZST.1.1_S20_L001_R1_001.fastq.gz

K.ZST.1.1	K.ZST.1.1_S20_L001_R2_001.fastq.gz
K.ZST.1.2	K.ZST.1.2_S21_L001_R1_001.fastq.gz
K.ZST.1.2	K.ZST.1.2_S21_L001_R2_001.fastq.gz
P.BeL.1.1	P.BeL.1.1_S22_L001_R1_001.fastq.gz
P.BeL.1.1	P.BeL.1.1_S22_L001_R2_001.fastq.gz
P.BeL.1.2	P.BeL.1.2_S23_L001_R1_001.fastq.gz
P.BeL.1.2	P.BeL.1.2_S23_L001_R2_001.fastq.gz
P.BeL.2.1	P.BeL.2.1_S24_L001_R1_001.fastq.gz
P.BeL.2.1	P.BeL.2.1_S24_L001_R2_001.fastq.gz
P.BeL.2.2	P.BeL.2.2_S25_L001_R1_001.fastq.gz
P.BeL.2.2	P.BeL.2.2_S25_L001_R2_001.fastq.gz
P.Bob.1.2	P.Bob.1.2_S26_L001_R1_001.fastq.gz
P.Bob.1.2	P.Bob.1.2_S26_L001_R2_001.fastq.gz
P.Ira	P.Ira_S27_L001_R1_001.fastq.gz
P.Ira	P.Ira_S27_L001_R2_001.fastq.gz
P.Ira.13.2	P.Ira.13.2_S28_L001_R1_001.fastq.gz
P.Ira.13.2	P.Ira.13.2_S28_L001_R2_001.fastq.gz
P.Ira.6	P.Ira.6_S29_L001_R1_001.fastq.gz
P.Ira.6	P.Ira.6_S29_L001_R2_001.fastq.gz
P.KKF.1.1	P.KKF.1.1_S30_L001_R1_001.fastq.gz
P.KKF.1.1	P.KKF.1.1_S30_L001_R2_001.fastq.gz
P.KKF.1.2	P.KKF.1.2_S31_L001_R1_001.fastq.gz
P.KKF.1.2	P.KKF.1.2_S31_L001_R2_001.fastq.gz
P.KKF.2.1	P.KKF.2.1_S32_L001_R1_001.fastq.gz
P.KKF.2.1	P.KKF.2.1_S32_L001_R2_001.fastq.gz
P.KKF.2.2	P.KKF.2.2_S33_L001_R1_001.fastq.gz
P.KKF.2.2	P.KKF.2.2_S33_L001_R2_001.fastq.gz
P.Leg.1.1	P.Leg.1.1_S34_L001_R1_001.fastq.gz
P.Leg.1.1	P.Leg.1.1_S34_L001_R2_001.fastq.gz
P.Leg.1.2	P.Leg.1.2_S35_L001_R1_001.fastq.gz
P.Leg.1.2	P.Leg.1.2_S35_L001_R2_001.fastq.gz
P.Leg.2.1	P.Leg.2.1_S36_L001_R1_001.fastq.gz
P.Leg.2.1	P.Leg.2.1_S36_L001_R2_001.fastq.gz
P.Leg.2.2	P.Leg.2.2_S37_L001_R1_001.fastq.gz
P.Leg.2.2	P.Leg.2.2_S37_L001_R2_001.fastq.gz
P.Leg.3.1	P.Leg.3.1_S38_L001_R1_001.fastq.gz
P.Leg.3.1	P.Leg.3.1_S38_L001_R2_001.fastq.gz
P.Leg.3.2	P.Leg.3.2_S39_L001_R1_001.fastq.gz
P.Leg.3.2	P.Leg.3.2_S39_L001_R2_001.fastq.gz
P.RM.2.1	P.RM.2.1_S40_L001_R1_001.fastq.gz
P.RM.2.1	P.RM.2.1_S40_L001_R2_001.fastq.gz

P.RM.2.2	P.RM.2.2_S41_L001_R1_001.fastq.gz
P.RM.2.2	P.RM.2.2_S41_L001_R2_001.fastq.gz
P.Rm.1.1	P.Rm.1.1_S42_L001_R1_001.fastq.gz
P.Rm.1.1	P.Rm.1.1_S42_L001_R2_001.fastq.gz
P.Rm.1.2	P.Rm.1.2_S43_L001_R1_001.fastq.gz
P.Rm.1.2	P.Rm.1.2_S43_L001_R2_001.fastq.gz
P.Rop	P.Rop_S44_L001_R1_001.fastq.gz
P.Rop	P.Rop_S44_L001_R2_001.fastq.gz
P.Rop.1.2	P.Rop.1.2_S45_L001_R1_001.fastq.gz
P.Rop.1.2	P.Rop.1.2_S45_L001_R2_001.fastq.gz
P.WJB.1.1	P.WJB.1.1_S46_L001_R1_001.fastq.gz
P.WJB.1.1	P.WJB.1.1_S46_L001_R2_001.fastq.gz
P.WJB.1.2	P.WJB.1.2_S47_L001_R1_001.fastq.gz
P.WJB.1.2	P.WJB.1.2_S47_L001_R2_001.fastq.gz
P.WRJ.1.1	P.WRJ.1.1_S48_L001_R1_001.fastq.gz
P.WRJ.1.1	P.WRJ.1.1_S48_L001_R2_001.fastq.gz
P.WRJ.1.2	P.WRJ.1.2_S49_L001_R1_001.fastq.gz
P.WRJ.1.2	P.WRJ.1.2_S49_L001_R2_001.fastq.gz
P.ZER.1.1	P.ZER.1.1_S50_L001_R1_001.fastq.gz
P.ZER.1.1	P.ZER.1.1_S50_L001_R2_001.fastq.gz
P.ZER.1.2	P.ZER.1.2_S51_L001_R1_001.fastq.gz
P.ZER.1.2	P.ZER.1.2_S51_L001_R2_001.fastq.gz
P.ZER.2.1	P.ZER.2.1_S52_L001_R1_001.fastq.gz
P.ZER.2.1	P.ZER.2.1_S52_L001_R2_001.fastq.gz
P.ZER.2.2	P.ZER.2.2_S53_L001_R1_001.fastq.gz
P.ZER.2.2	P.ZER.2.2_S53_L001_R2_001.fastq.gz

Installed QIIME2 plugins

plugin	version
alignment	2026.7.0
boots	2026.7.0
composition	2026.7.0
cutadapt	2026.7.0
dada2	2026.7.0
deblur	2026.7.0
demux	2026.7.0
diversity	2026.7.0
diversity-lib	2026.7.0
emperor	2026.7.0
feature-classifier	2026.7.0
feature-table	2026.7.0
fondue	2026.7.0
fragment-insertion	2026.7.0
itsxpress	2.2.0
kmerizer	2026.7.0
longitudinal	2026.7.0
metadata	2026.7.0
phylogeny	2026.7.0
quality-control	2026.7.0
quality-filter	2026.7.0
rescript	2026.7.0
sample-classifier	2026.7.0
stats	2026.7.0
taxa	2026.7.0
types	2026.7.0
vizard	2026.7.0
vsearch	2026.7.0

DADA2 read retention

DADA2 denoises raw reads into exact amplicon sequence variants (ASVs) and removes chimeric artifacts. This table shows what fraction of each sample's reads survived quality filtering, denoising, merging (paired-end only), and chimera removal -- a sample retaining very few reads at any step may need different DADA2 parameters, or exclusion from downstream analysis.

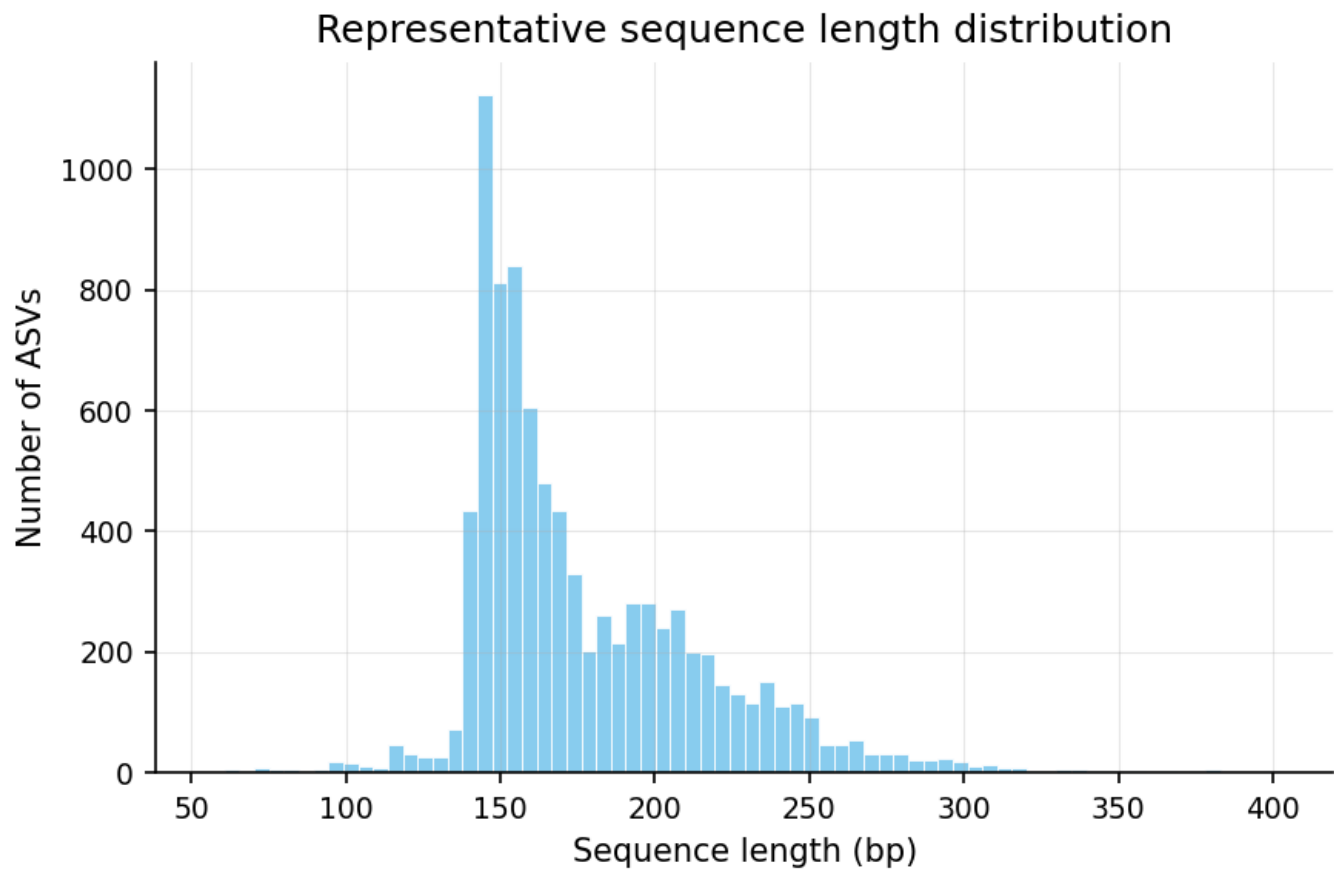
sample	input	filtered	% passed filter	denoised	merged	% merged	non-chimeric	% non-chimeric
K.BeL.1.1	186097.0	118644.0	63.8	118343.0	110929.0	59.6	110406.0	59.3
K.BeL.1.2	173700.0	125167.0	72.1	124756.0	122074.0	70.3	120481.0	69.4
K.CSL.1.1	78936.0	67182.0	85.1	66809.0	63994.0	81.1	63543.0	80.5
K.CSL.1.2	112897.0	90753.0	80.4	90258.0	87373.0	77.4	86669.0	76.8
K.CT9.1.1	130636.0	113776.0	87.1	113297.0	112414.0	86.0	112250.0	85.9
K.CtE.1.2	83746.0	72356.0	86.4	72082.0	69220.0	82.7	68972.0	82.4
K.Ira.13.2	83729.0	69567.0	83.1	69523.0	69331.0	82.8	69116.0	82.5
K.KKF.1.1	88091.0	76044.0	86.3	75842.0	74835.0	85.0	74654.0	84.8
K.KKF.1.2	244671.0	205490.0	84.0	205072.0	200110.0	81.8	199415.0	81.5
K.KKF.2.1	75123.0	63323.0	84.3	62744.0	60900.0	81.1	60423.0	80.4
K.KKF.2.2	113190.0	93511.0	82.6	92788.0	90727.0	80.2	90224.0	79.7
K.Leg.1.1	79526.0	41366.0	52.0	41118.0	39579.0	49.8	39579.0	49.8
K.Leg.1.2	84574.0	70488.0	83.3	70051.0	68106.0	80.5	67881.0	80.3
K.WKJ.2.1	98900.0	86643.0	87.6	86485.0	85270.0	86.2	83410.0	84.3
K.WKJ.2.2	112999.0	93963.0	83.2	93686.0	91362.0	80.8	87081.0	77.1
K.ZER.1.1	98842.0	76634.0	77.5	76227.0	74805.0	75.7	72964.0	73.8
K.ZER.1.2	121339.0	92921.0	76.6	92394.0	91205.0	75.2	90188.0	74.3
K.ZER.2.1	92514.0	81241.0	87.8	80726.0	79142.0	85.5	79061.0	85.5
K.ZER.2.2	125357.0	106606.0	85.0	106103.0	104038.0	83.0	103808.0	82.8
K.ZST.1.1	121584.0	92949.0	76.5	92181.0	86091.0	70.8	85779.0	70.5
K.ZST.1.2	104092.0	75603.0	72.6	74874.0	71075.0	68.3	70753.0	68.0
P.BeL.1.1	270479.0	212814.0	78.7	212435.0	211446.0	78.2	209017.0	77.3
P.BeL.1.2	261439.0	185348.0	70.9	185063.0	184205.0	70.5	181725.0	69.5

P.BeL.2.1	216111.0	165372.0	76.5	165202.0	164564.0	76.2	164550.0	76.1
P.BeL.2.2	191815.0	153049.0	79.8	152750.0	151867.0	79.2	151534.0	79.0
P.Bob.1.2	440543.0	366131.0	83.1	365734.0	361695.0	82.1	359729.0	81.7
P.Ira	193299.0	155369.0	80.4	155156.0	153196.0	79.2	152490.0	78.9
P.Ira.13.2	263735.0	205612.0	78.0	205202.0	203877.0	77.3	203877.0	77.3
P.Ira.6	80462.0	58531.0	72.7	58499.0	57848.0	71.9	57527.0	71.5
P.KKF.1.1	114636.0	104024.0	90.7	103941.0	103669.0	90.4	103437.0	90.2
P.KKF.1.2	102006.0	91611.0	89.8	91450.0	91145.0	89.3	91041.0	89.2
P.KKF.2.1	132540.0	122687.0	92.6	122432.0	121852.0	91.9	121845.0	91.9
P.KKF.2.2	144428.0	134078.0	92.8	134002.0	133642.0	92.5	133543.0	92.5
P.Leg.1.1	479204.0	393923.0	82.2	393550.0	390341.0	81.5	389161.0	81.2
P.Leg.1.2	127551.0	110189.0	86.4	109981.0	108955.0	85.4	108918.0	85.4
P.Leg.2.1	122651.0	97888.0	79.8	97265.0	93720.0	76.4	93386.0	76.1
P.Leg.2.2	73323.0	57207.0	78.0	56774.0	55756.0	76.0	55476.0	75.7
P.Leg.3.1	56994.0	28247.0	49.6	28061.0	26010.0	45.6	25999.0	45.6
P.Leg.3.2	94091.0	73306.0	77.9	73165.0	63857.0	67.9	63718.0	67.7
P.Rm.1.1	144413.0	118954.0	82.4	118835.0	116382.0	80.6	115323.0	79.9
P.Rm.1.2	135992.0	115795.0	85.2	115710.0	114226.0	84.0	113761.0	83.7
P.RM.2.1	66585.0	56596.0	85.0	56480.0	55393.0	83.2	55393.0	83.2
P.RM.2.2	92260.0	57898.0	62.8	57831.0	52297.0	56.7	52088.0	56.5
P.Rop	239643.0	172157.0	71.8	171736.0	170350.0	71.1	170105.0	71.0
P.Rop.1.2	388389.0	315494.0	81.2	314931.0	284535.0	73.3	282561.0	72.8
P.WJB.1.1	81090.0	69568.0	85.8	69134.0	67793.0	83.6	66944.0	82.6
P.WJB.1.2	113221.0	99466.0	87.8	98991.0	97427.0	86.0	96945.0	85.6
P.WRJ.1.1	69656.0	53361.0	76.6	53130.0	52469.0	75.3	52240.0	75.0
P.WRJ.1.2	114966.0	104661.0	91.0	104510.0	103594.0	90.1	103594.0	90.1
P.ZER.1.1	82316.0	58191.0	70.7	58003.0	56878.0	69.1	56878.0	69.1
P.ZER.1.2	164181.0	130150.0	79.3	130004.0	127163.0	77.5	127163.0	77.5

P.ZER.2.1	164623.0	131449.0	79.8	131360.0	128958.0	78.3	128952.0	78.3
P.ZER.2.2	180337.0	143925.0	79.8	143821.0	140393.0	77.8	140393.0	77.8

Sequence length distribution

Length distribution of the representative sequences (one per ASV) produced by DADA2. ITS regions are naturally variable in length, but a distribution that is unexpectedly narrow, wide, or off-target suggests the primer/extraction step included flanking regions it should not have, or trimmed too aggressively.



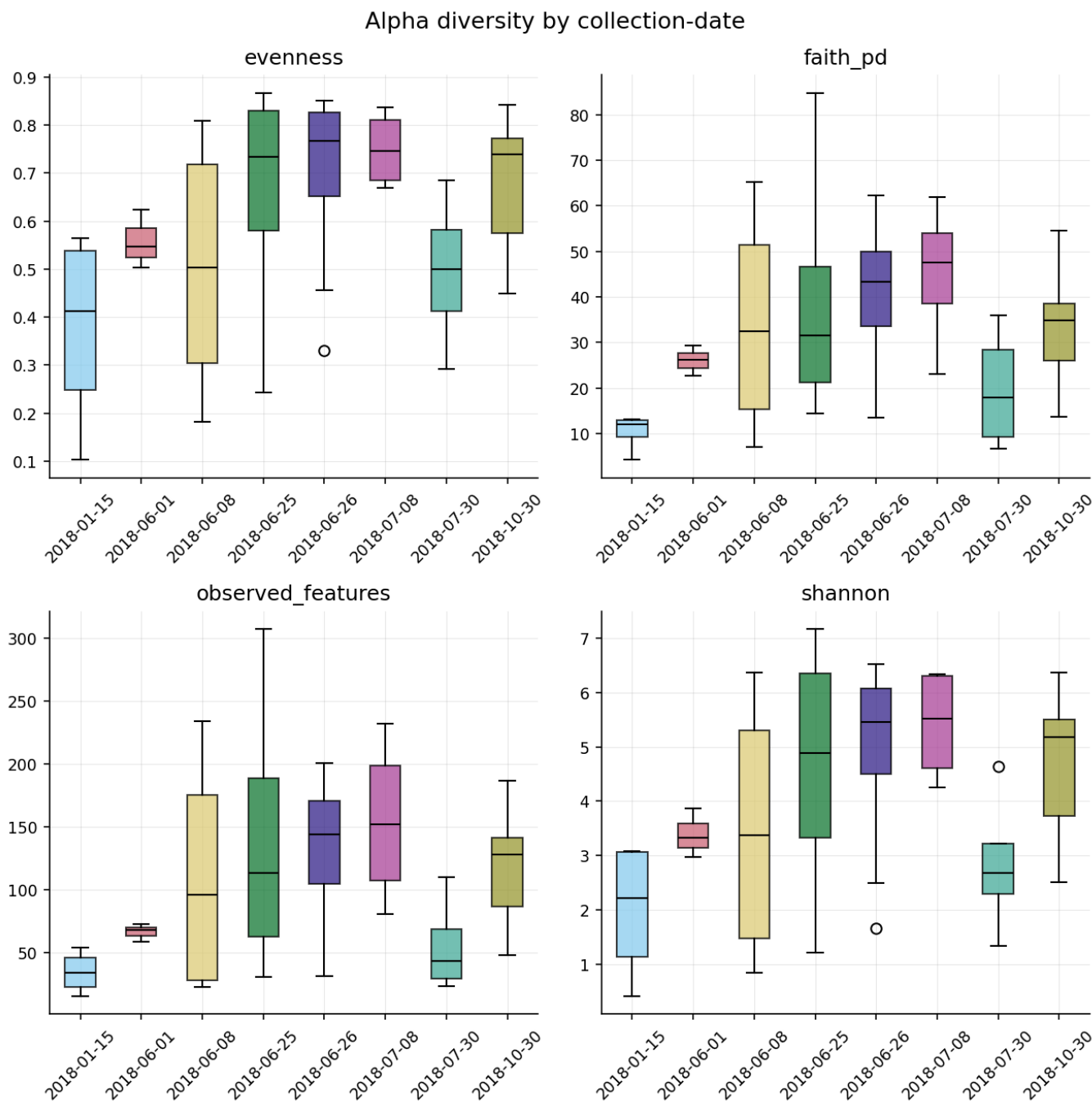
Alpha diversity group significance (Kruskal-Wallis)

Alpha diversity measures within-sample richness and evenness of taxa (Faith's phylogenetic diversity, observed features, Shannon, and evenness). A Kruskal-Wallis test checks whether a metric differs significantly between the groups of a metadata column; a low *p*-value (conventionally < 0.05) indicates the groups differ.

metric	metadata column	H	p-value
evenness	replicate	0.008	0.929
evenness	collection-date	12.900	0.075
evenness	env-local-scale	25.213	0.014
evenness	country	4.500	0.034
evenness	env-medium	29.044	0.000
faith_pd	replicate	0.182	0.669
faith_pd	collection-date	14.638	0.041
faith_pd	env-local-scale	31.146	0.002
faith_pd	country	8.981	0.003
faith_pd	env-medium	24.937	0.000
observed_features	replicate	0.103	0.749
observed_features	collection-date	13.583	0.059
observed_features	env-local-scale	33.029	0.001
observed_features	country	6.464	0.011
observed_features	env-medium	28.549	0.000
shannon	replicate	0.081	0.776
shannon	collection-date	13.065	0.071
shannon	env-local-scale	28.559	0.005
shannon	country	4.939	0.026
shannon	env-medium	29.784	0.000

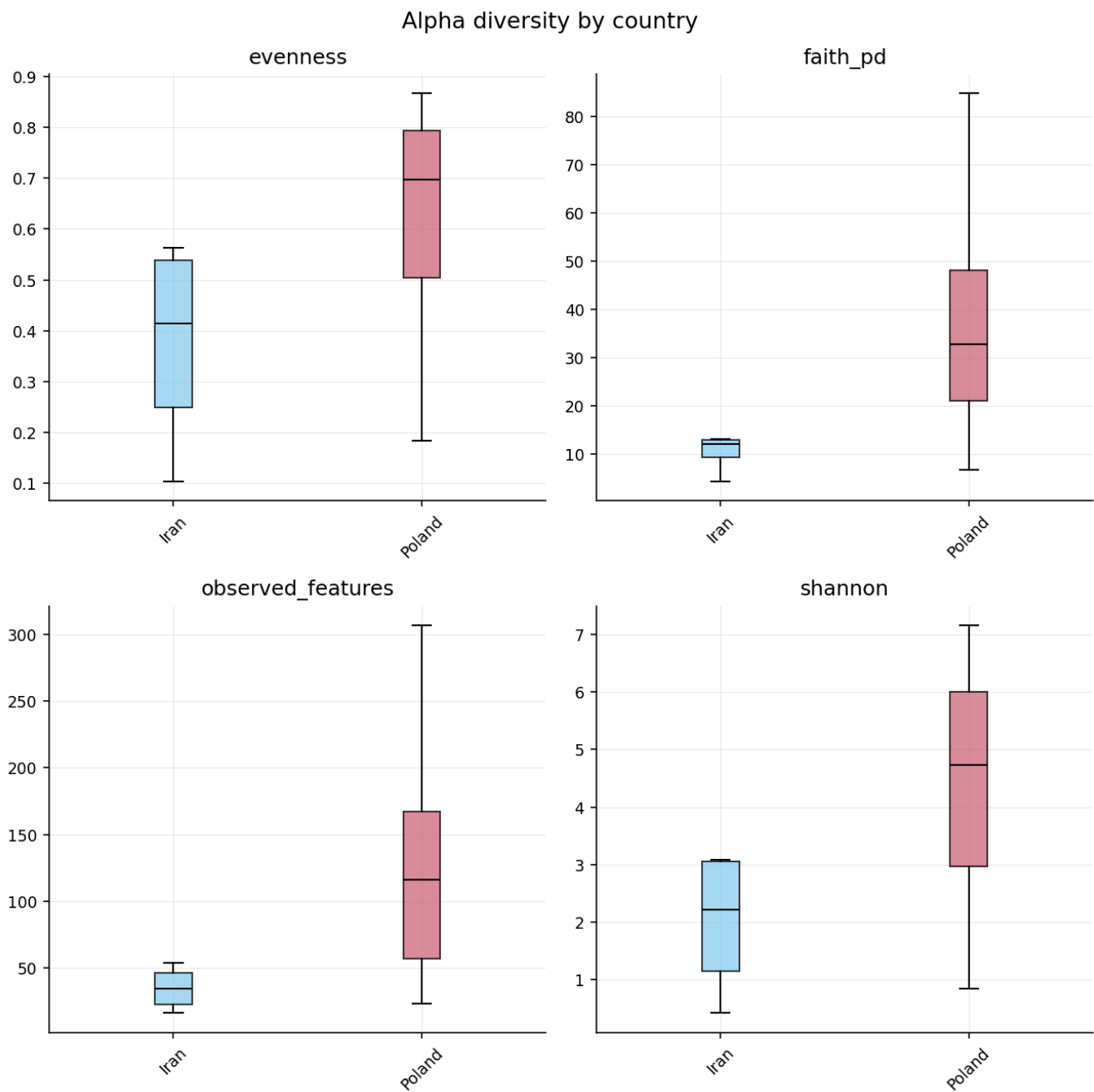
Alpha diversity by collection-date

Boxplots of each alpha diversity metric, split by this metadata column -- the same comparison as the group-significance table above, shown visually. Shown here because collection-date came back statistically significant ($p < 0.05$) above.



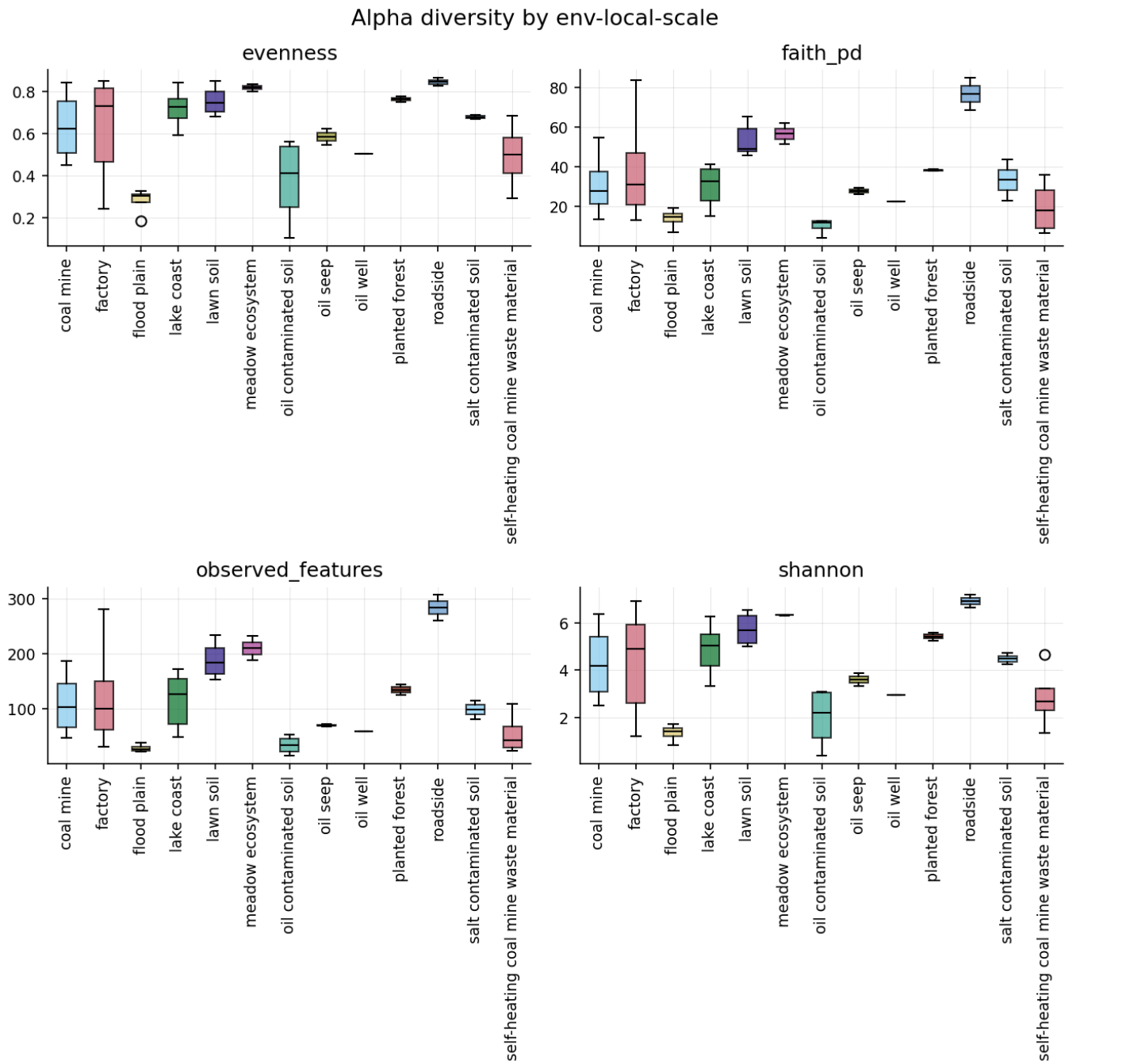
Alpha diversity by country

Boxplots of each alpha diversity metric, split by this metadata column -- the same comparison as the group-significance table above, shown visually. Shown here because country came back statistically significant ($p < 0.05$) above.



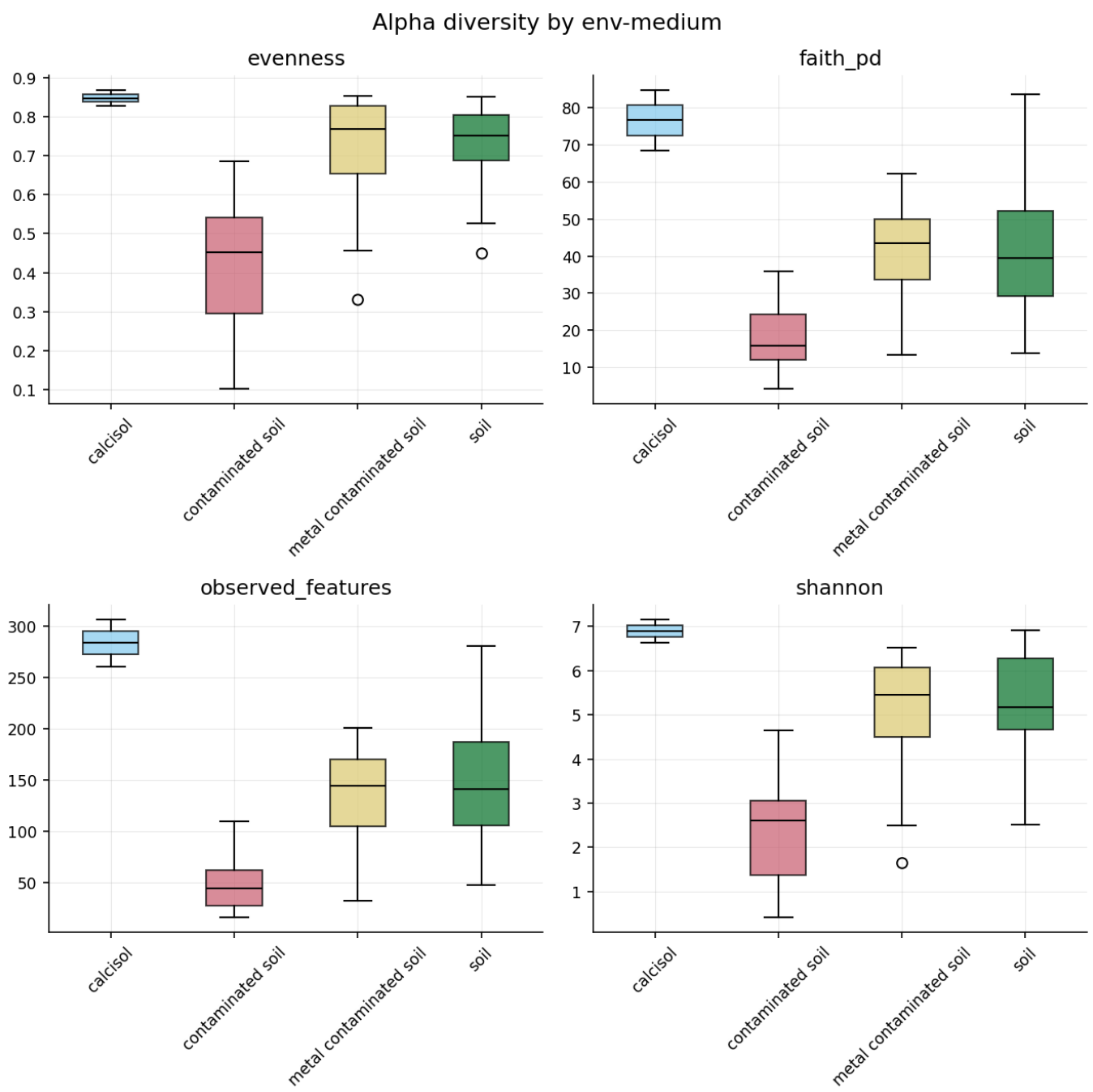
Alpha diversity by env-local-scale

Boxplots of each alpha diversity metric, split by this metadata column -- the same comparison as the group-significance table above, shown visually. Shown here because env-local-scale came back statistically significant ($p < 0.05$) above.



Alpha diversity by env-medium

Boxplots of each alpha diversity metric, split by this metadata column -- the same comparison as the group-significance table above, shown visually. Shown here because env-medium came back statistically significant ($p < 0.05$) above.

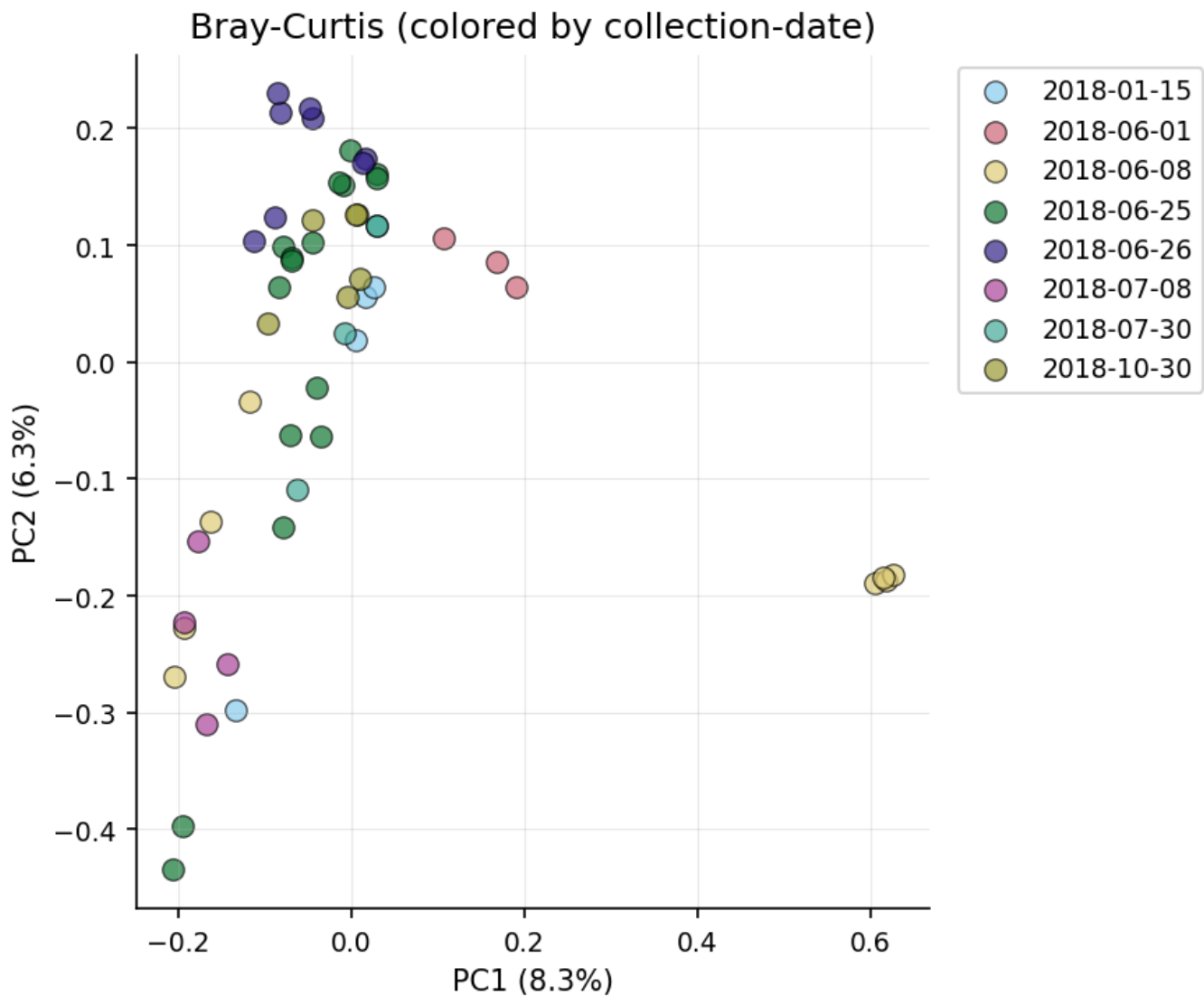


Beta diversity group significance (PERMANOVA)

Beta diversity compares community composition (not just richness) between samples. PERMANOVA tests whether samples within the same metadata group are more similar to each other than to samples in other groups, for both Bray-Curtis (abundance-weighted) and unweighted UniFrac (phylogenetic presence/absence) distances.

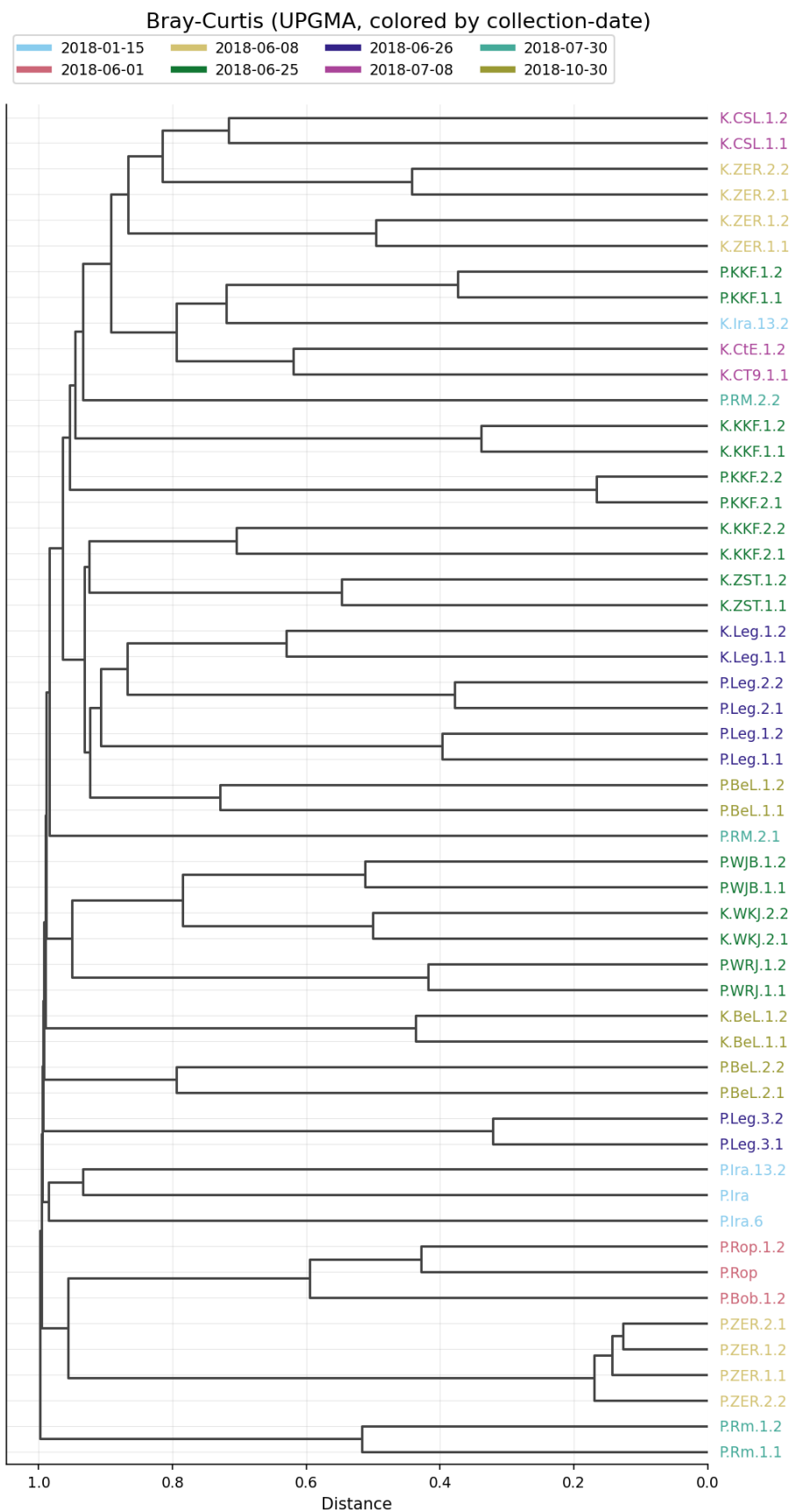
metadata column	distance metric	statistic	value	p-value
collection-date	bray_curtis	pseudo-F	2.324	0.001
collection-date	unweighted_unifrac	pseudo-F	2.215	0.001
country	bray_curtis	pseudo-F	1.338	0.061
country	unweighted_unifrac	pseudo-F	2.621	0.001
env-local-scale	bray_curtis	pseudo-F	2.508	0.001
env-local-scale	unweighted_unifrac	pseudo-F	2.313	0.001
env-medium	bray_curtis	pseudo-F	2.329	0.001
env-medium	unweighted_unifrac	pseudo-F	2.854	0.001
replicate	bray_curtis	pseudo-F	0.374	1.000
replicate	unweighted_unifrac	pseudo-F	0.558	1.000

Principal coordinates analysis (PCoA) projects the pairwise distance matrix into two dimensions for visualization; samples that cluster together have more similar community composition. The percentage on each axis is the proportion of total variance it explains. Shown here because collection-date came back statistically significant ($p < 0.05$) above.



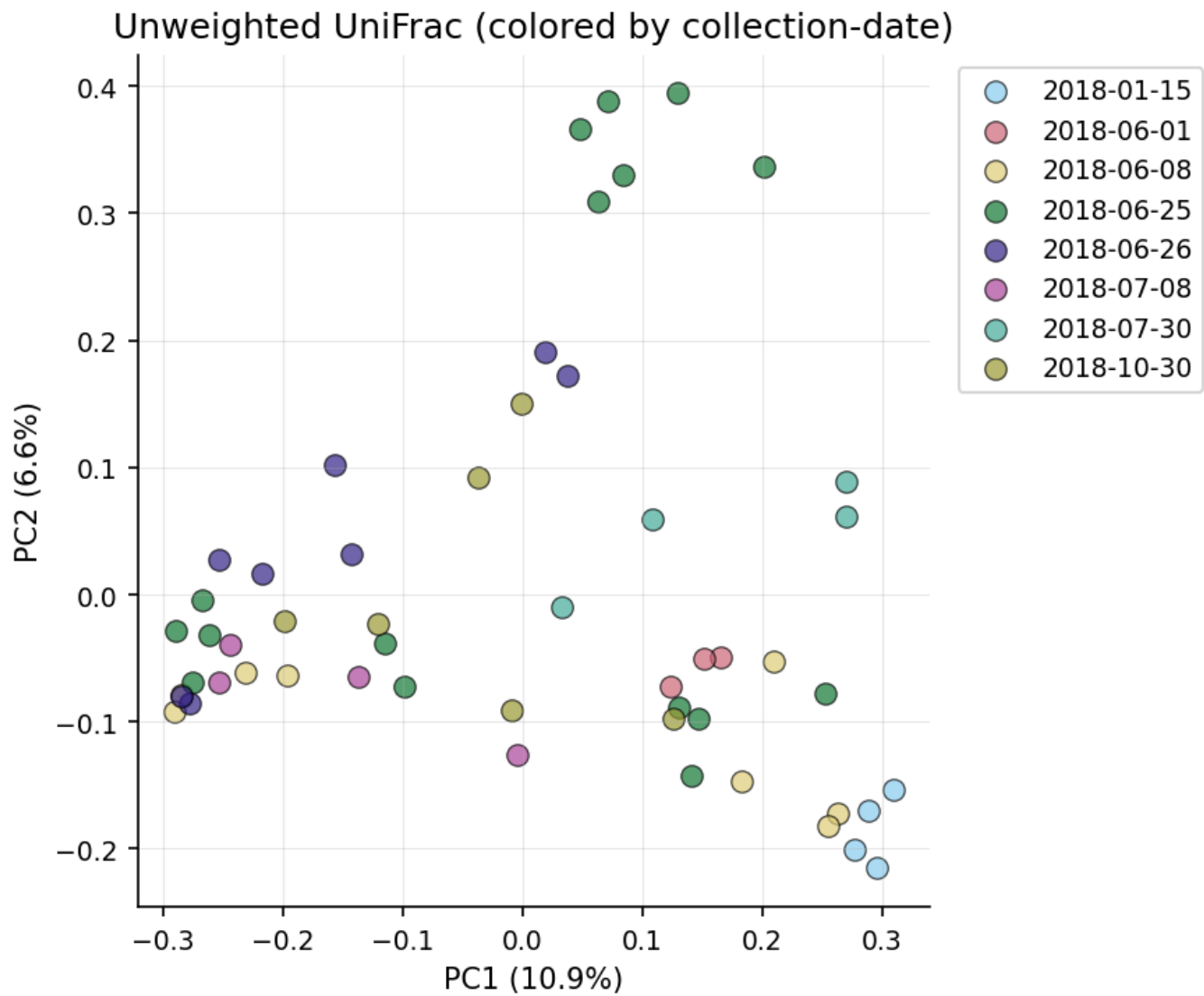
collection-date: Bray-Curtis sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because collection-date came back statistically significant ($p < 0.05$) above.



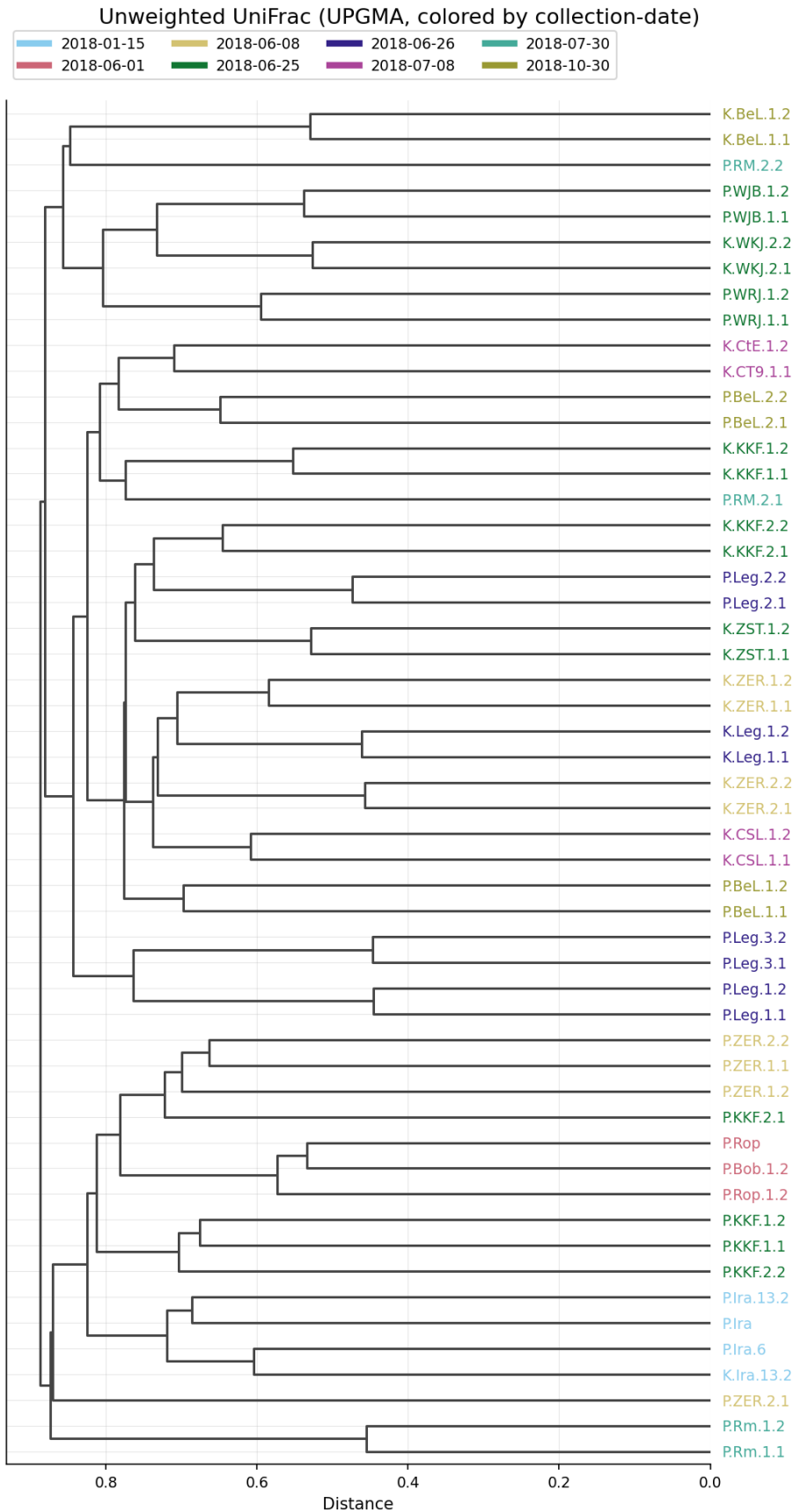
collection-date: Unweighted UniFrac PCoA

Principal coordinates analysis (PCoA) projects the pairwise distance matrix into two dimensions for visualization; samples that cluster together have more similar community composition. The percentage on each axis is the proportion of total variance it explains. Shown here because collection-date came back statistically significant ($p < 0.05$) above.



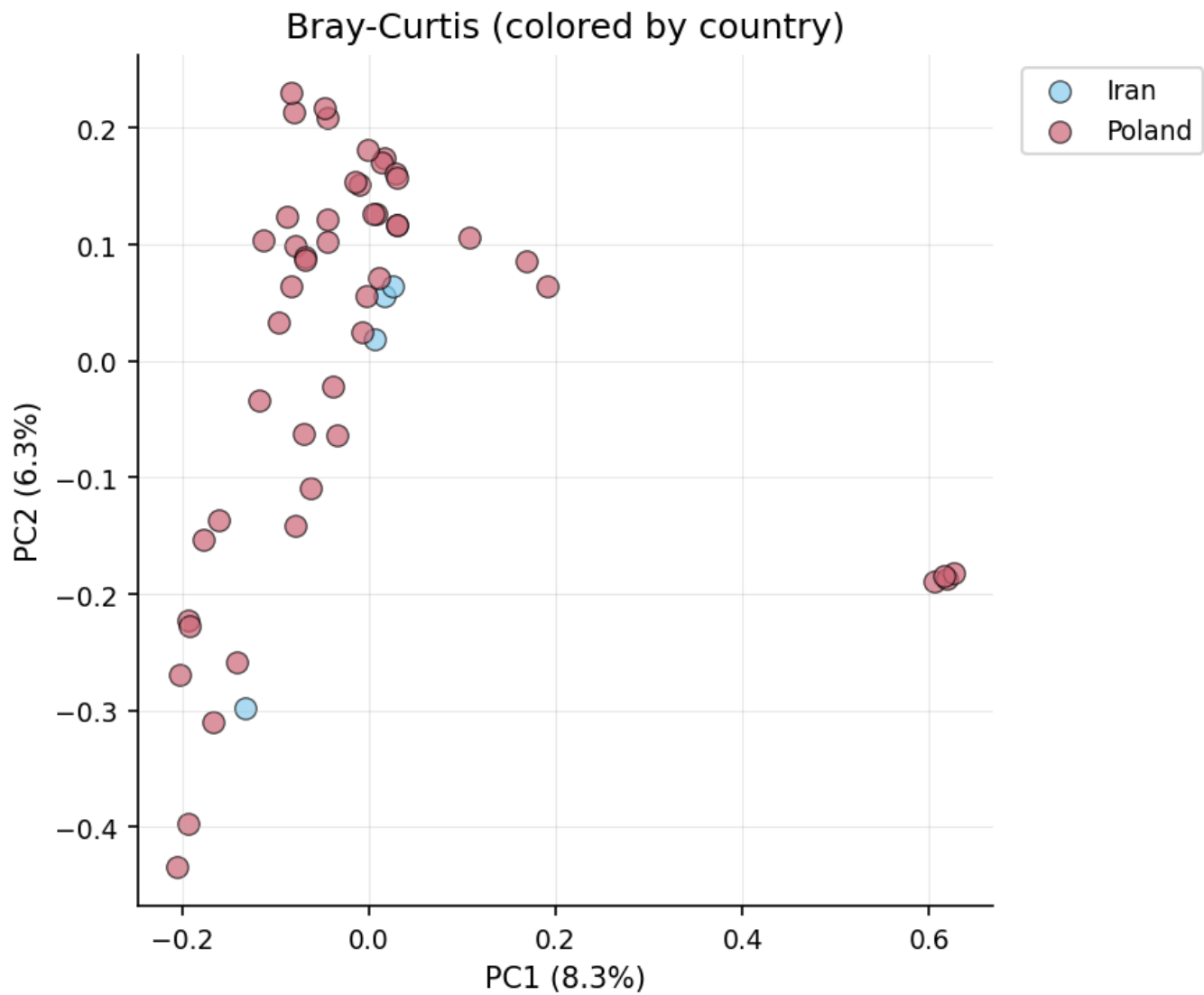
collection-date: Unweighted UniFrac sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because collection-date came back statistically significant ($p < 0.05$) above.



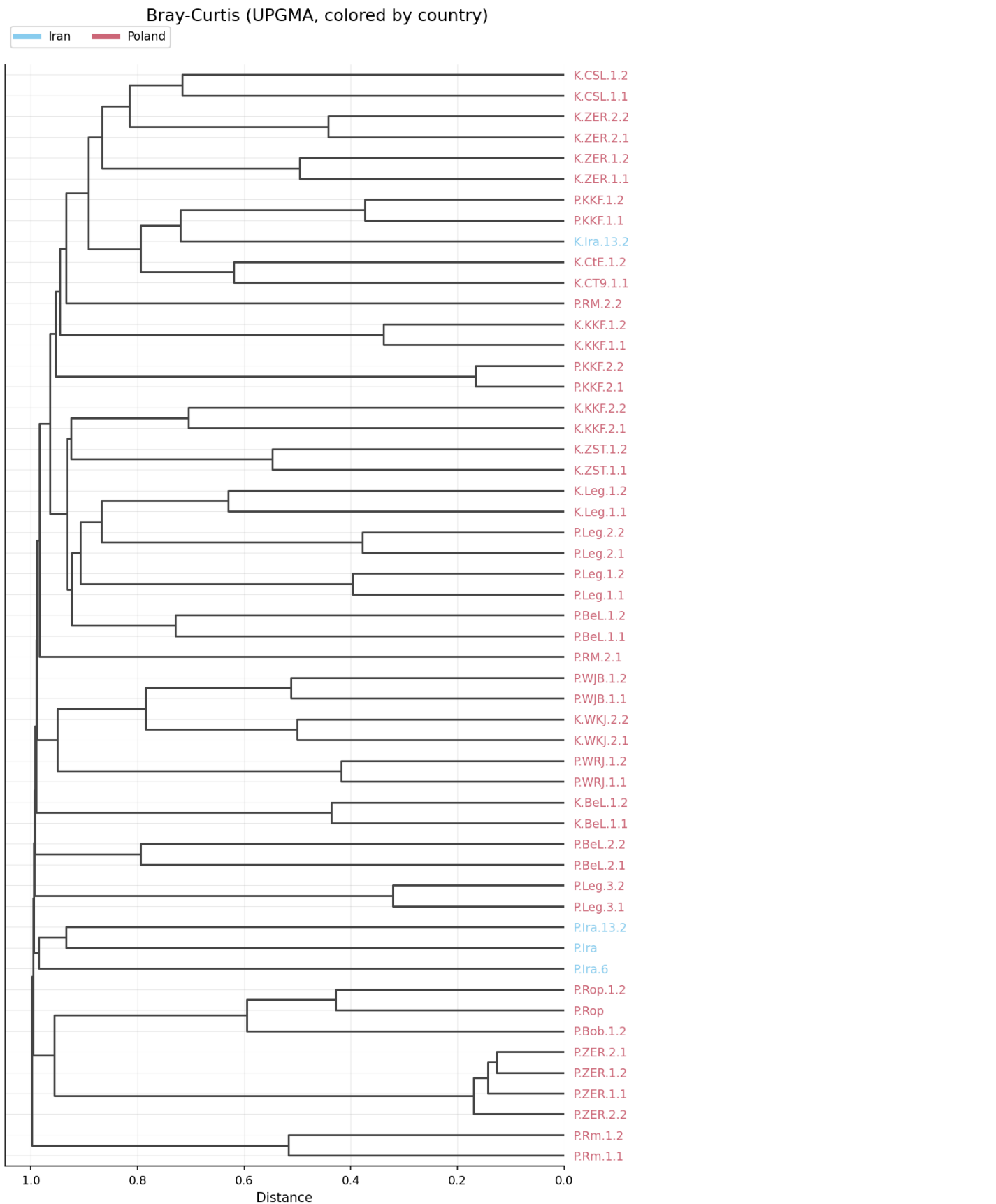
country: Bray-Curtis PCoA

Principal coordinates analysis (PCoA) projects the pairwise distance matrix into two dimensions for visualization; samples that cluster together have more similar community composition. The percentage on each axis is the proportion of total variance it explains. Shown here because country came back statistically significant ($p < 0.05$) above.



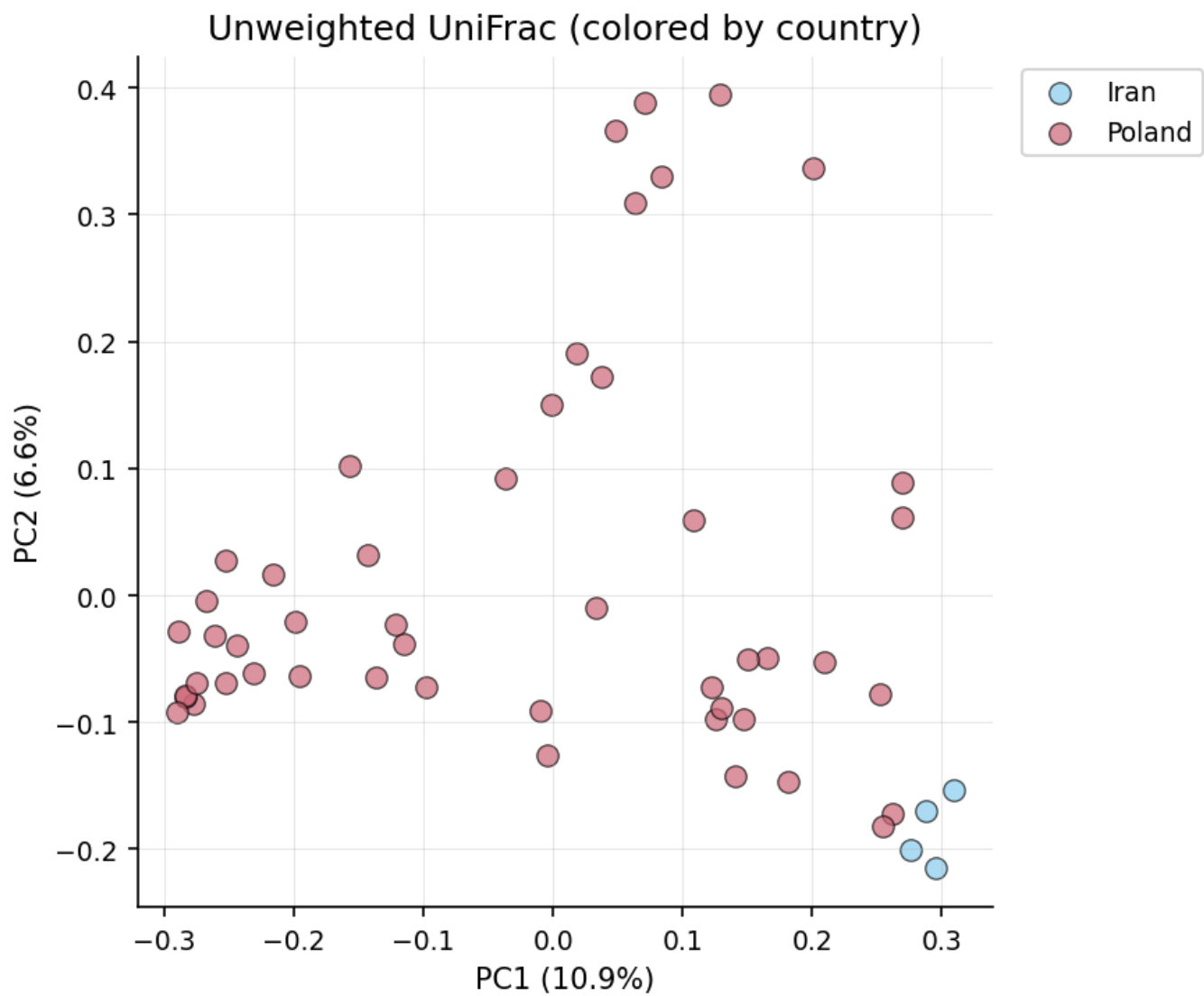
country: Bray-Curtis sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because country came back statistically significant ($p < 0.05$) above.



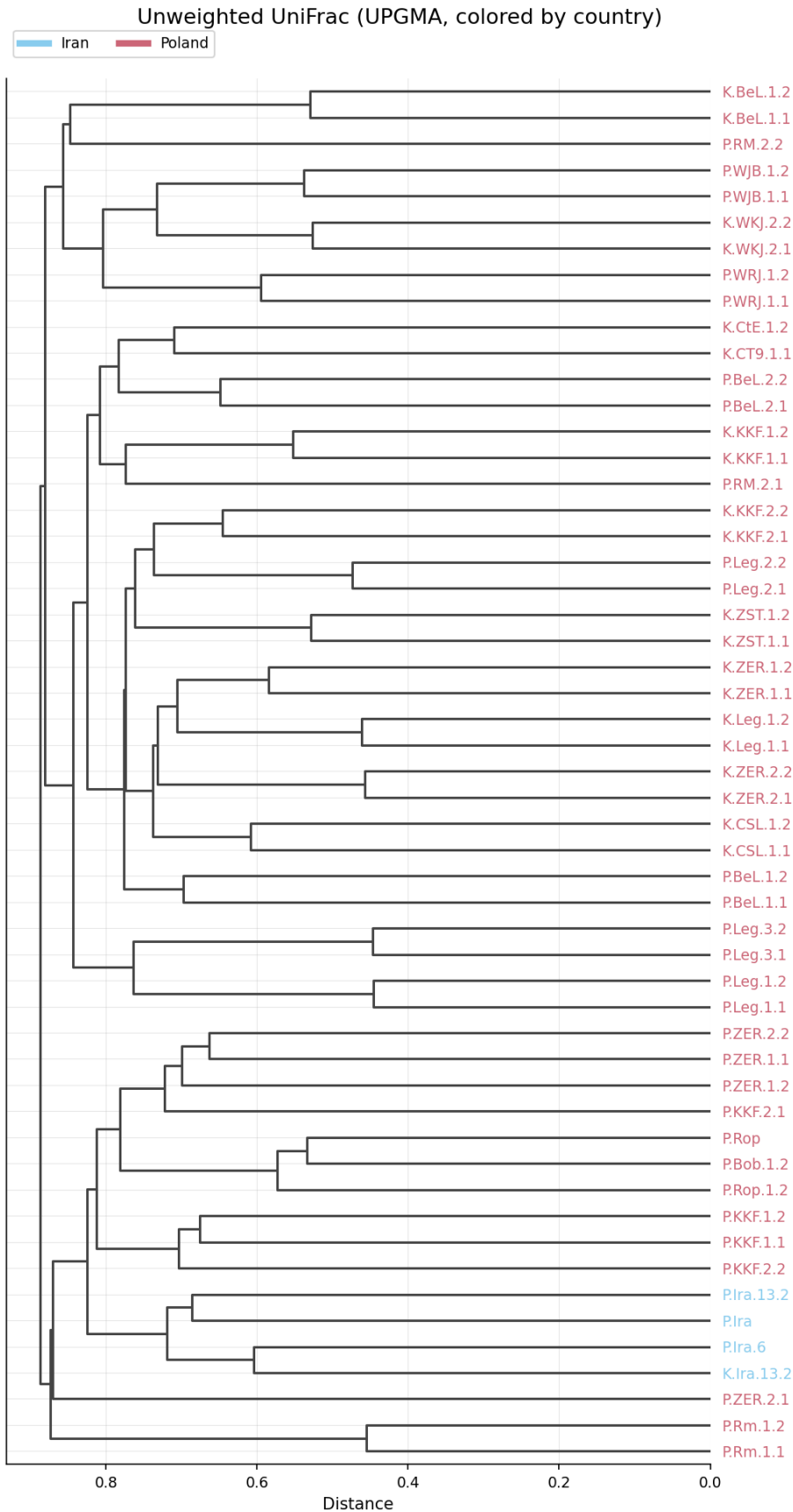
country: Unweighted UniFrac PCoA

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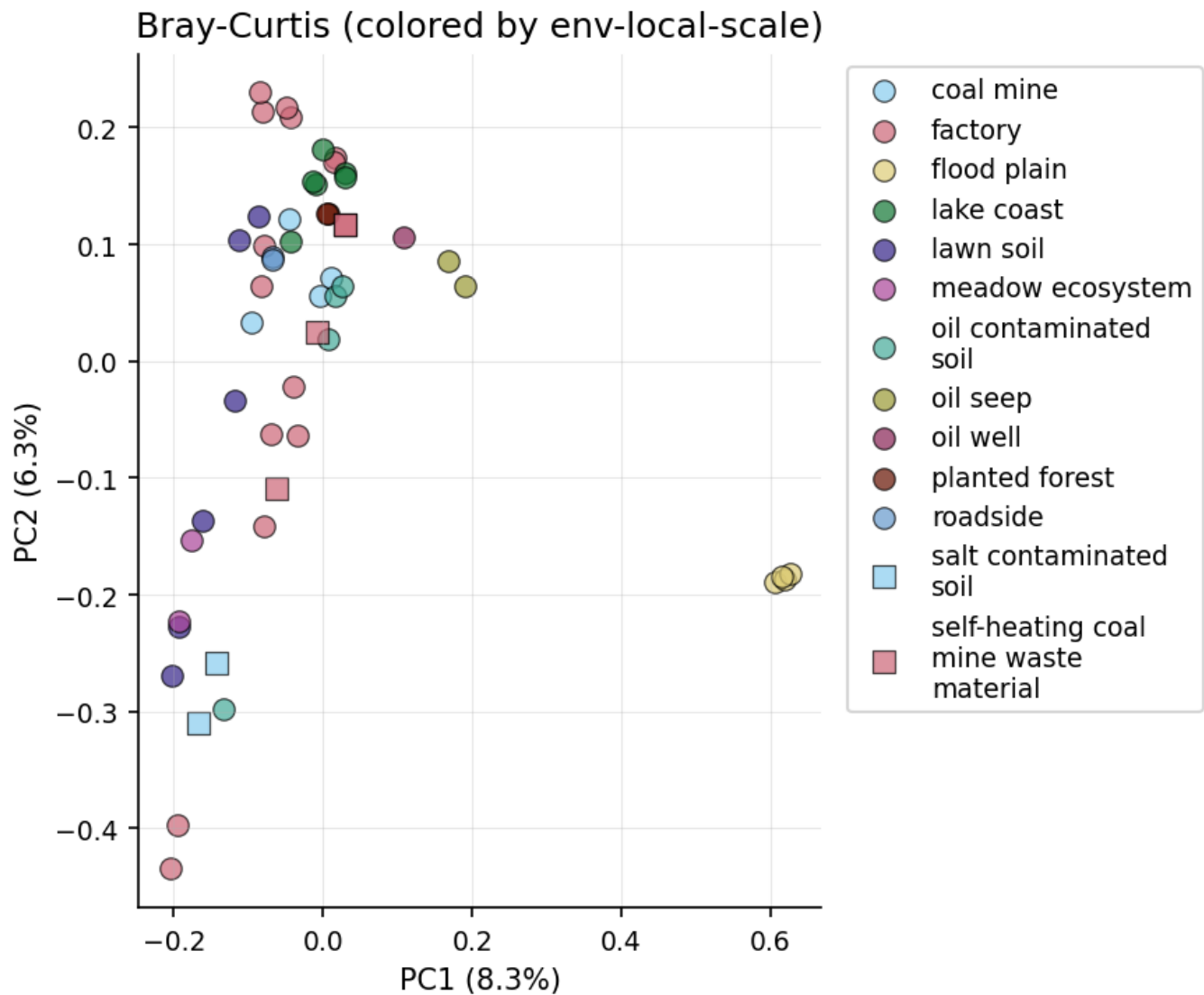
country: Unweighted UniFrac sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because country came back statistically significant ($p < 0.05$) above.



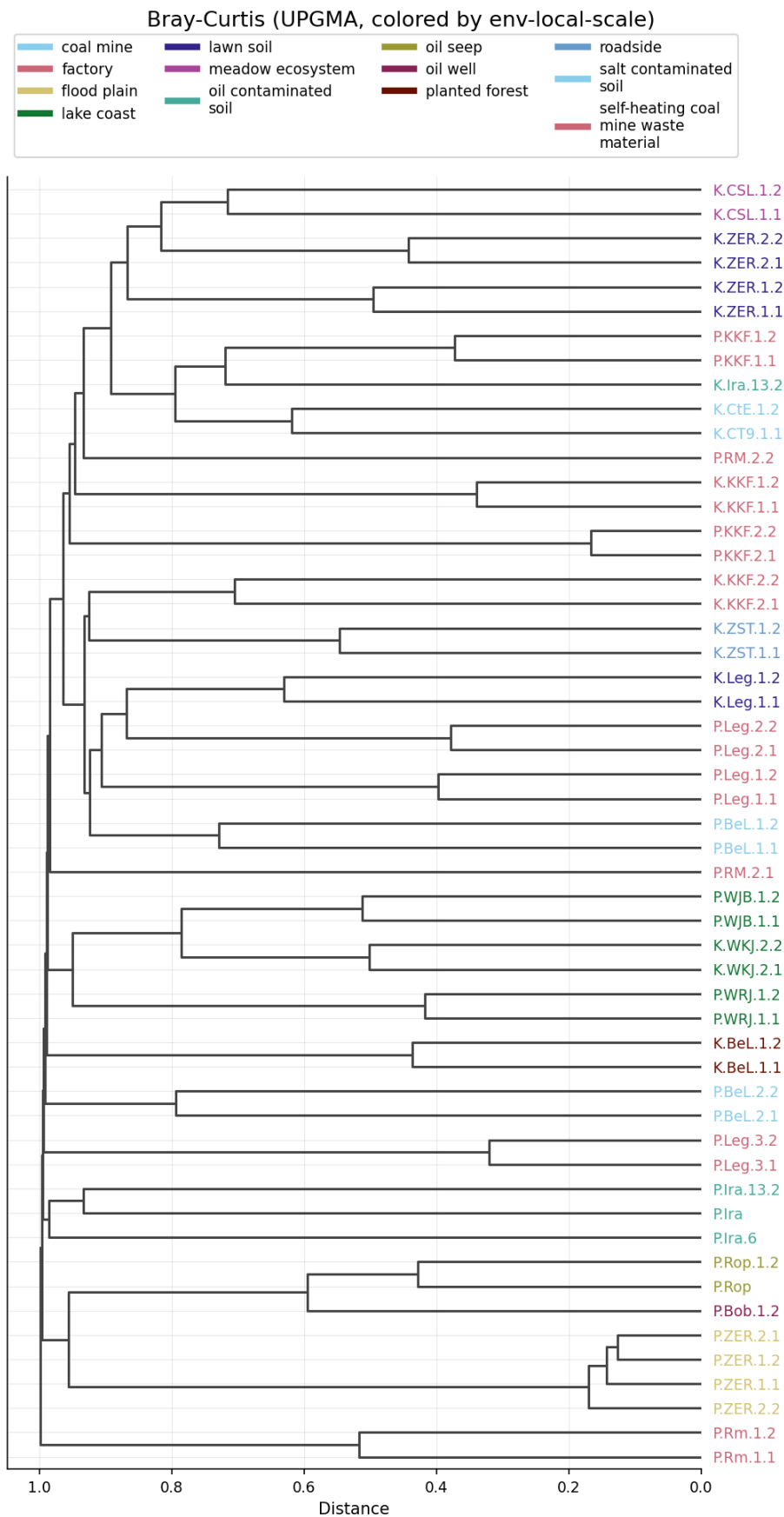
env-local-scale: Bray-Curtis PCoA

Principal coordinates analysis (PCoA) projects the pairwise distance matrix into two dimensions for visualization; samples that cluster together have more similar community composition. The percentage on each axis is the proportion of total variance it explains. Marker shape also distinguishes groups that share a color (this column has more groups than the palette has colors). Shown here because env-local-scale came back statistically significant ($p < 0.05$) above.



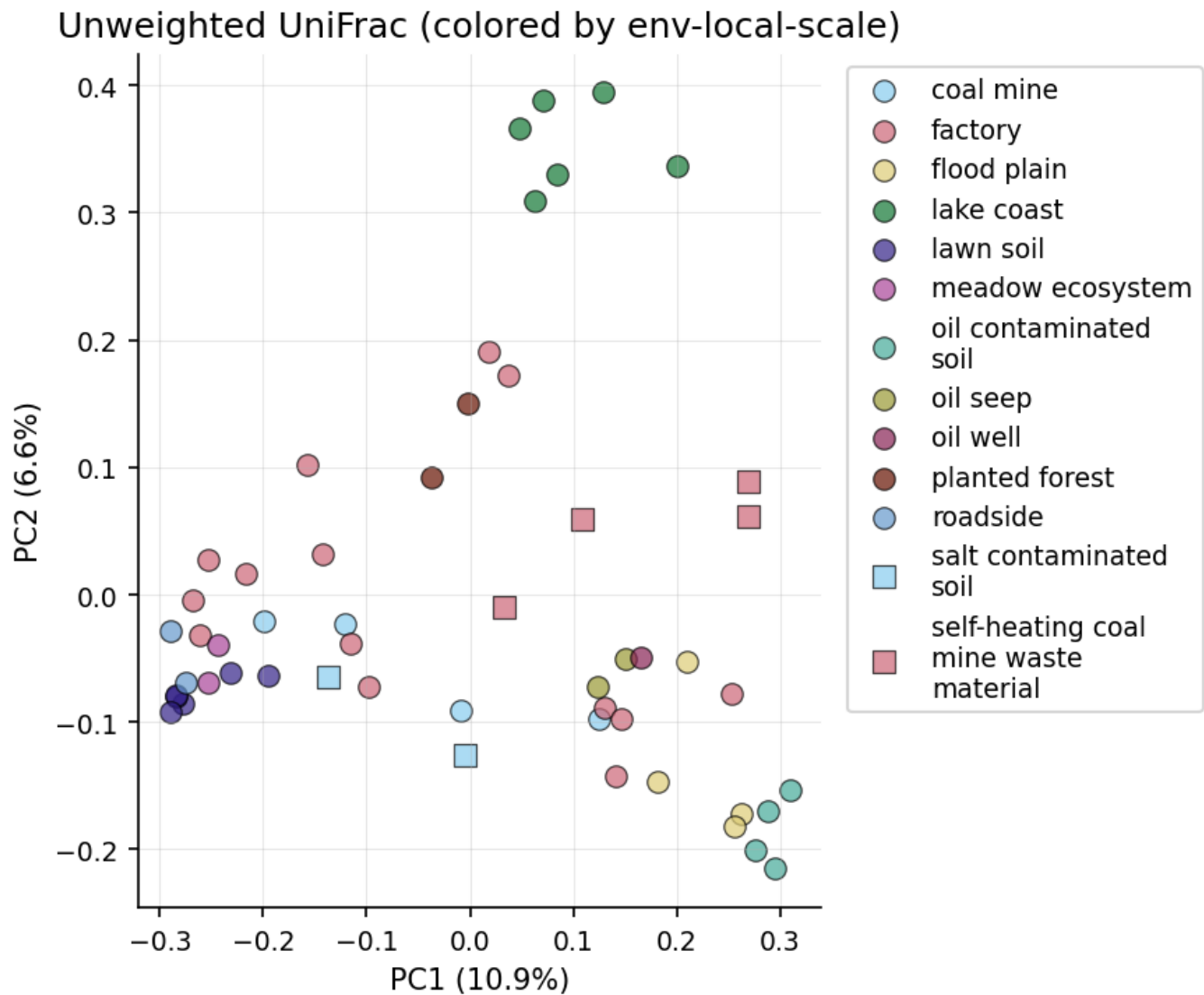
env-local-scale: Bray-Curtis sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because env-local-scale came back statistically significant ($p < 0.05$) above.



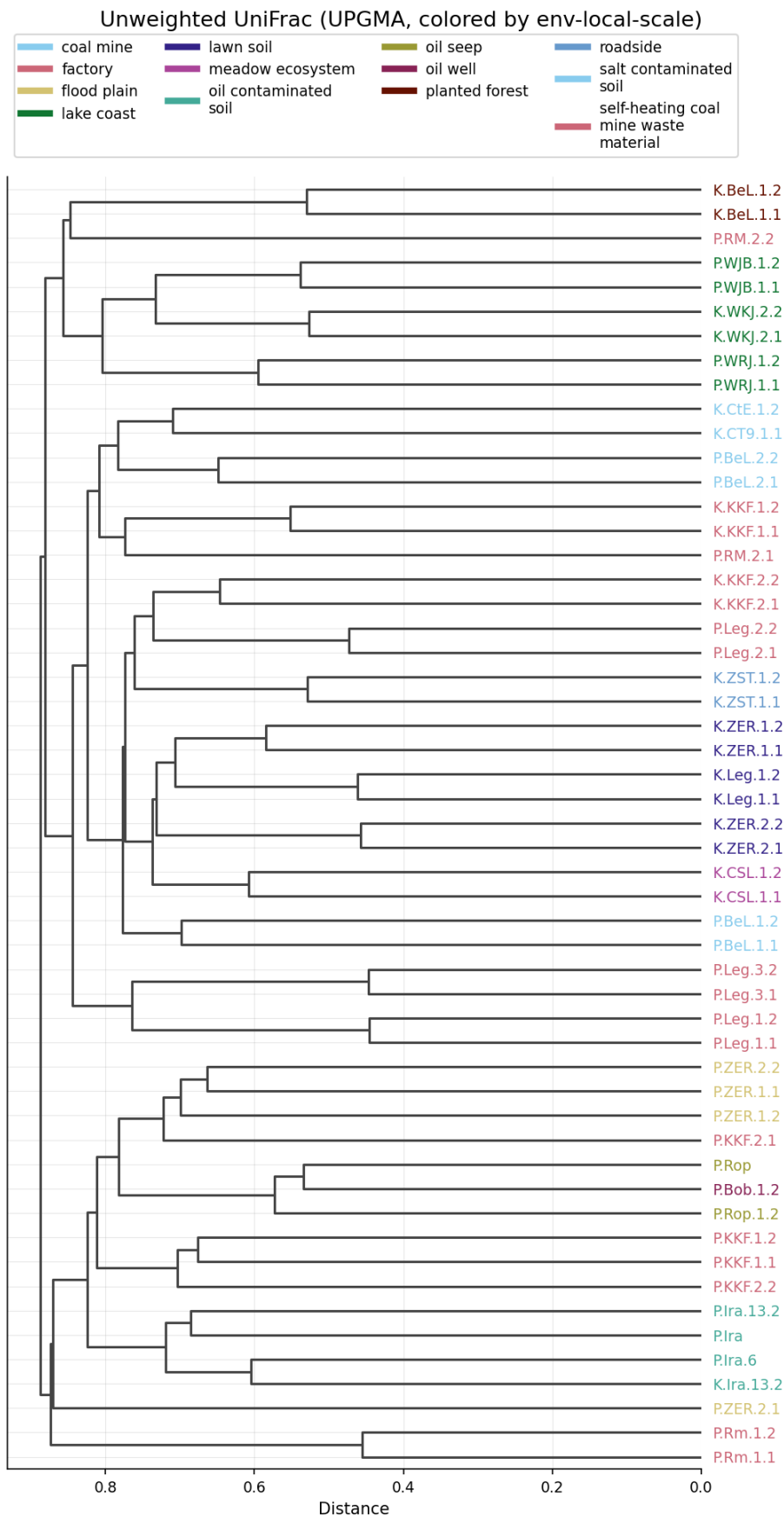
env-local-scale: Unweighted UniFrac PCoA

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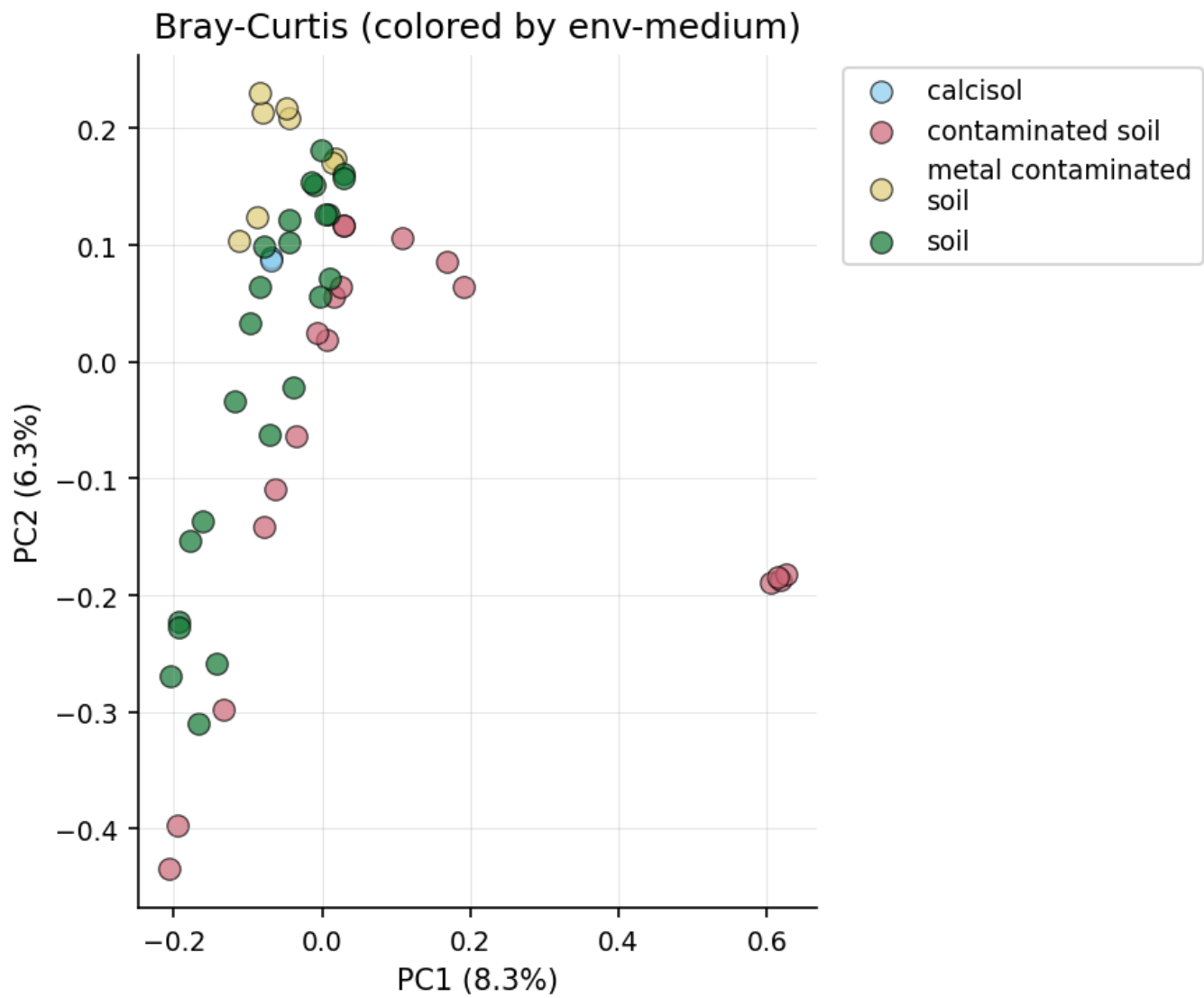
env-local-scale: Unweighted UniFrac sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because env-local-scale came back statistically significant ($p < 0.05$) above.



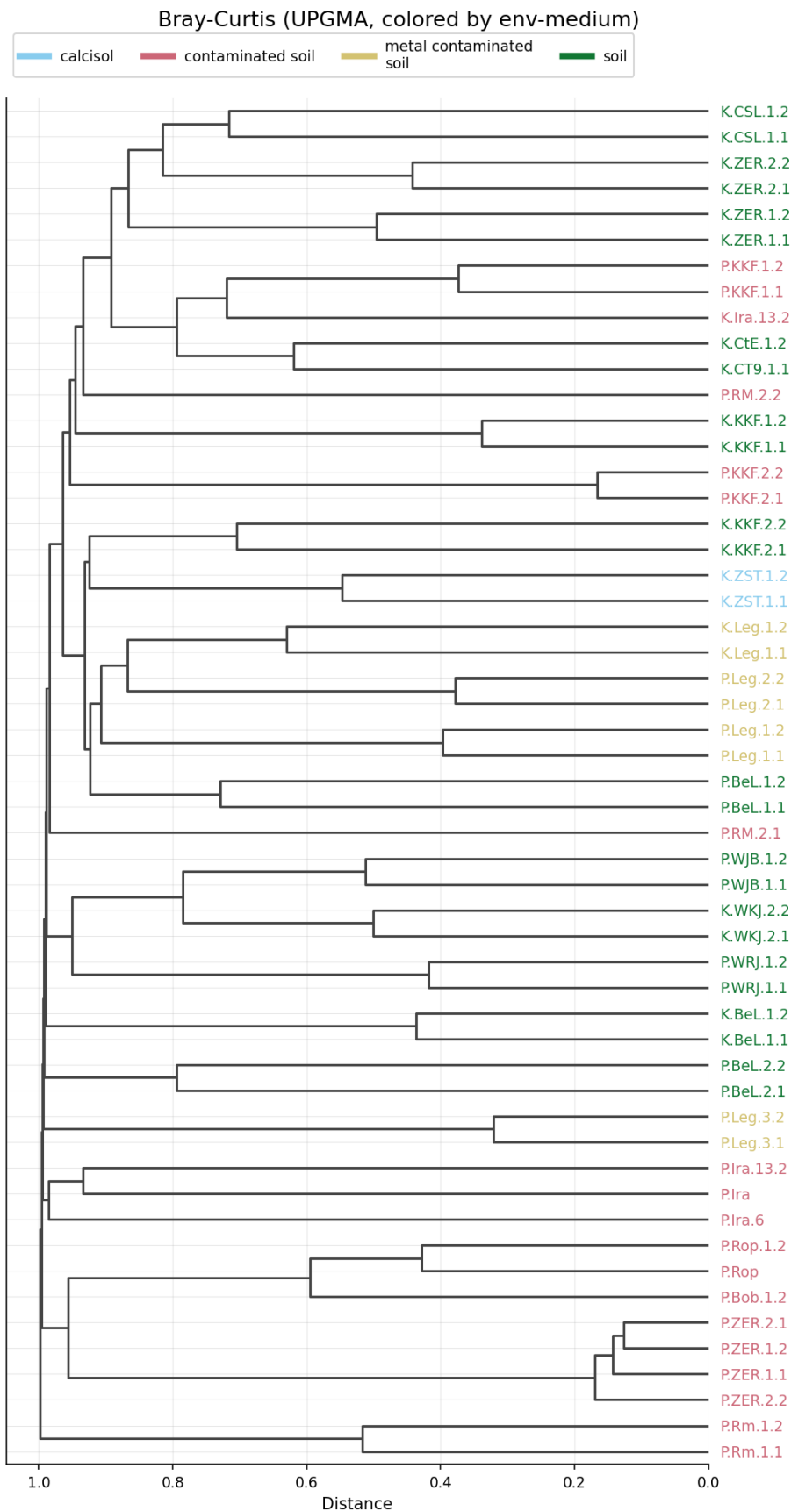
env-medium: Bray-Curtis PCoA

Principal coordinates analysis (PCoA) projects the pairwise distance matrix into two dimensions for visualization; samples that cluster together have more similar community composition. The percentage on each axis is the proportion of total variance it explains. Shown here because env-medium came back statistically significant ($p < 0.05$) above.



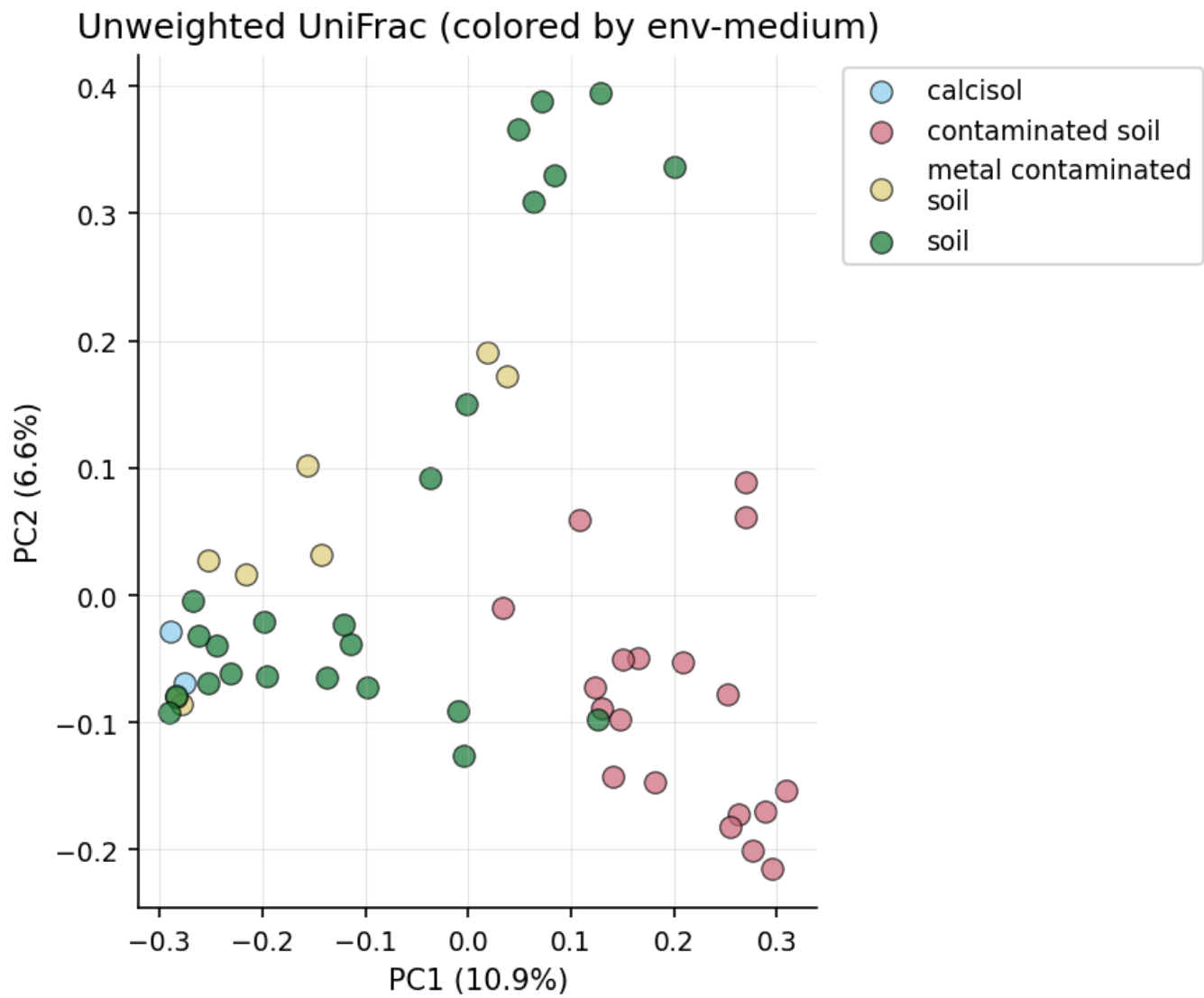
env-medium: Bray-Curtis sample clustering

UPGMA hierarchical clustering of samples from the same pairwise distance matrix used for the PCoA above; samples that join at a shorter distance are more similar in community composition. Unlike PCoA, this preserves the exact pairwise distances rather than a two-dimensional approximation of them. Shown here because env-medium came back statistically significant ($p < 0.05$) above.



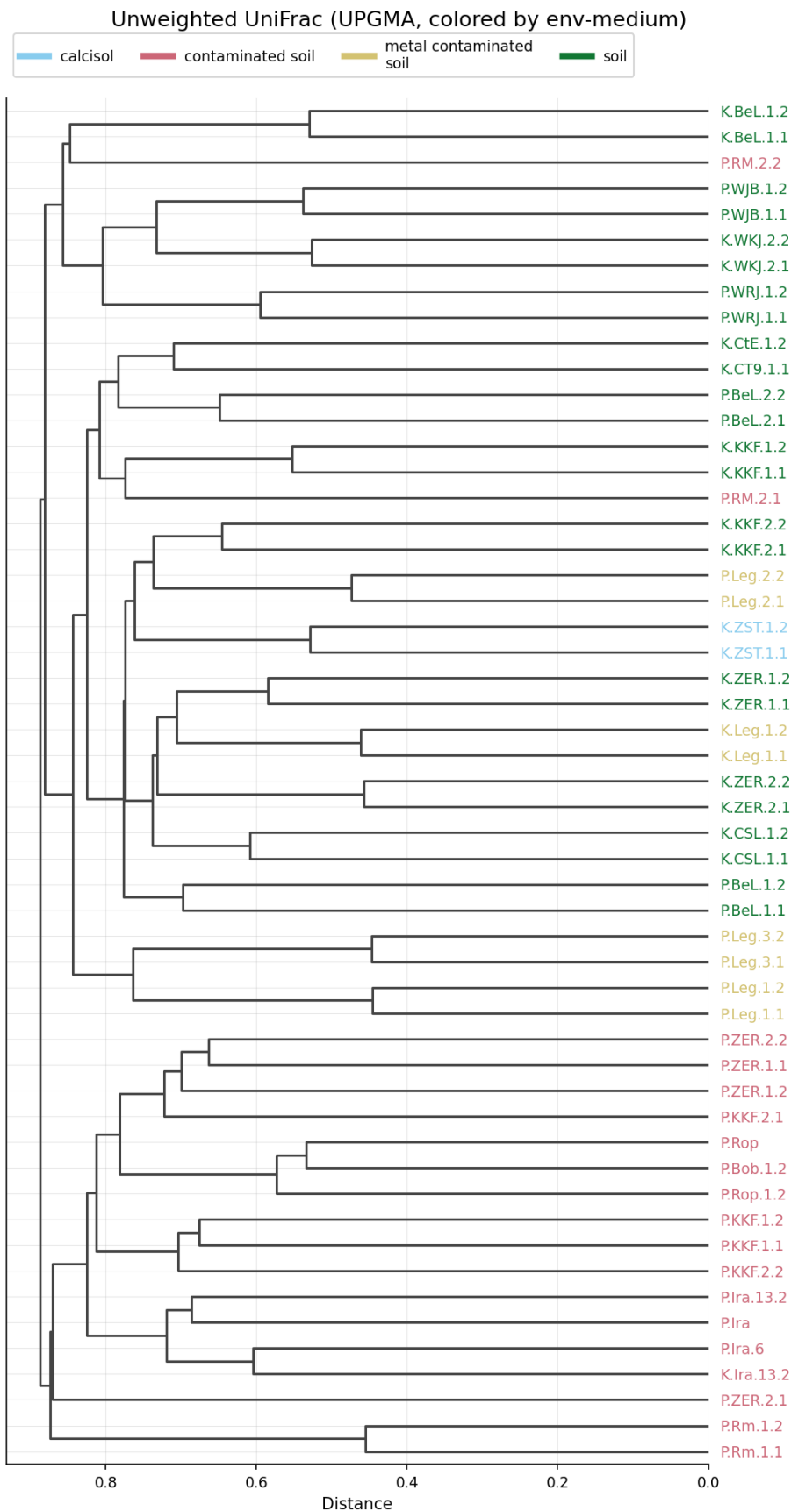
env-medium: Unweighted UniFrac PCoA

Principal coordinates analysis (PCoA) projects the pairwise distance matrix into two dimensions for visualization; samples that cluster together have more similar community composition. The percentage on each axis is the proportion of total variance it explains. Shown here because env-medium came back statistically significant ($p < 0.05$) above.



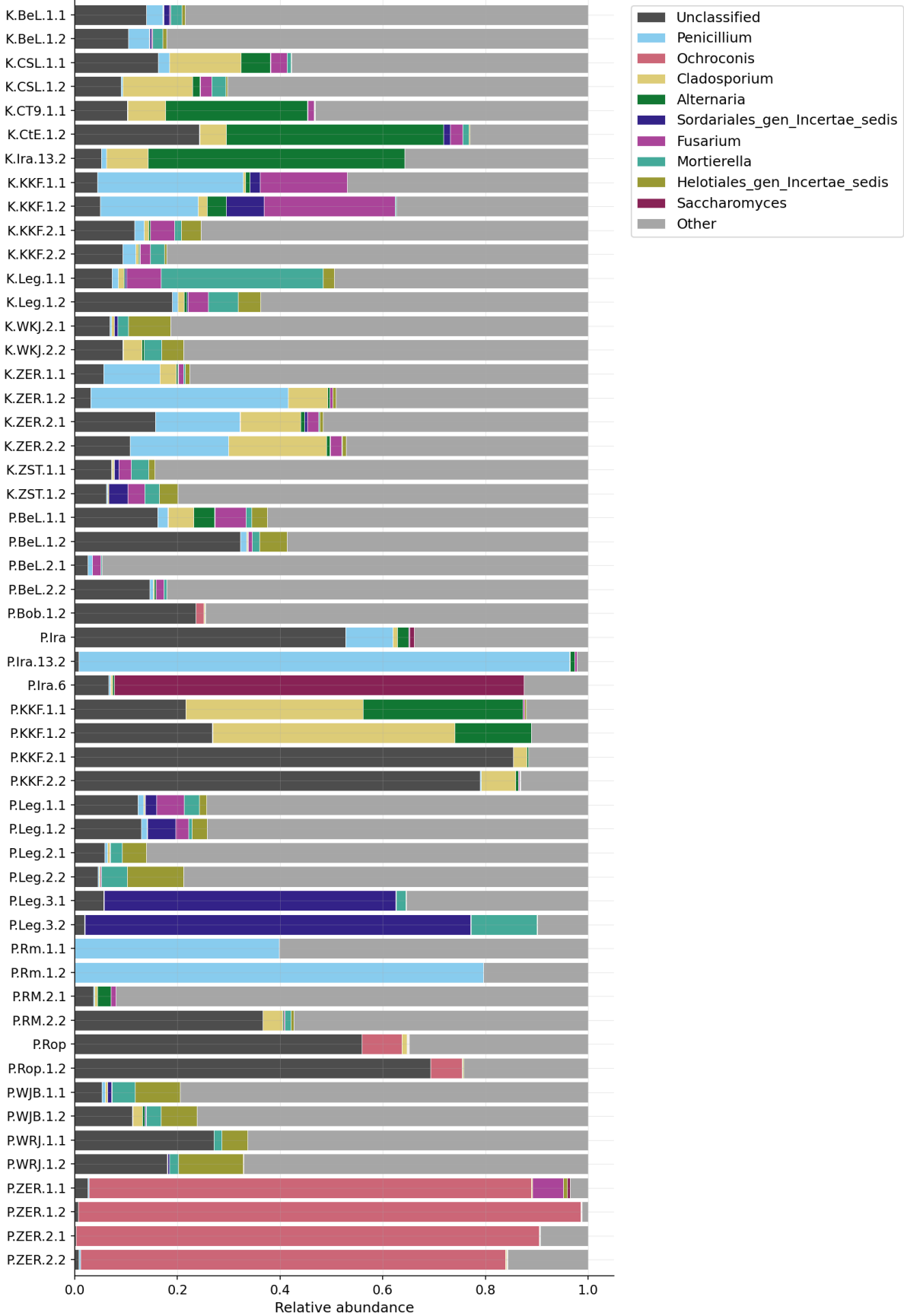
env-medium: Unweighted UniFrac sample clustering

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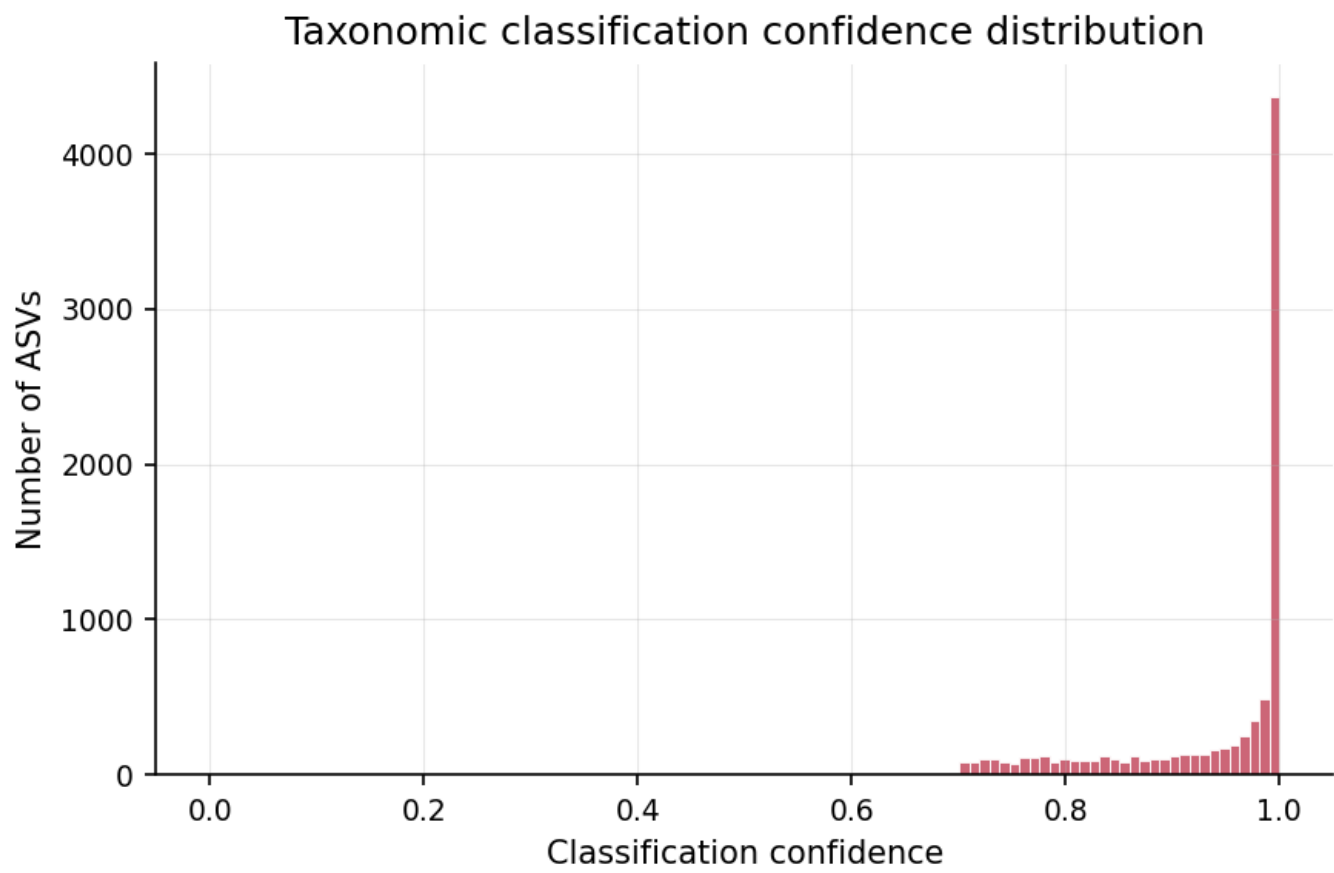
Genus-level relative abundance

Relative abundance of the most abundant genera in each sample, collapsed from the classifier's taxonomic assignments. "Unclassified" covers features the classifier could not assign to a genus; "Other" covers genera outside the most abundant ones shown individually.



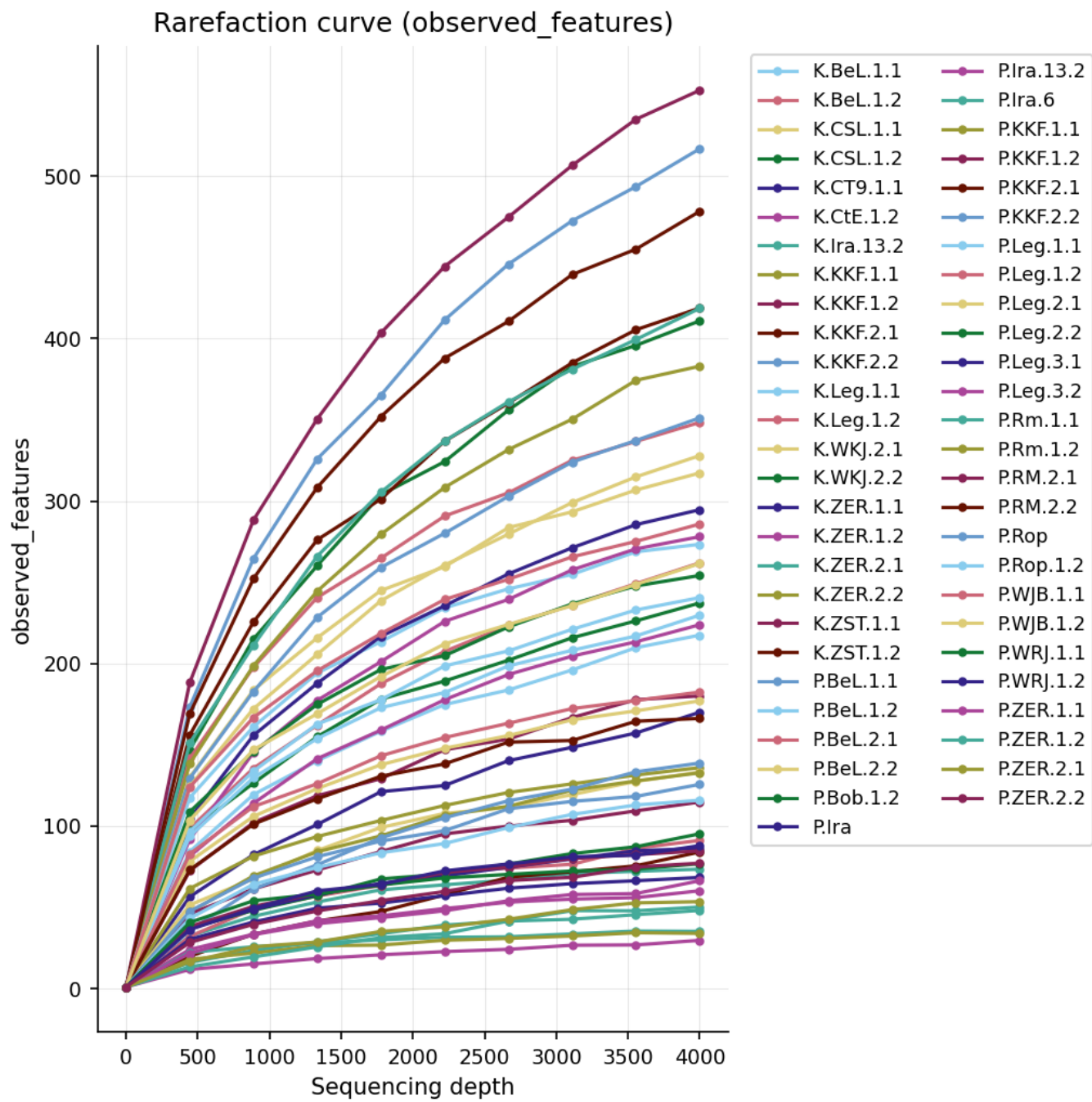
Classification confidence

Classifier confidence score for each ASV's taxonomic assignment (1.0 is fully confident). A distribution skewed toward low values suggests many ASVs are only weakly resolved by the reference database used to train the classifier.



Rarefaction curve

Number of observed features (ASVs) as a function of sequencing depth, per sample. A curve that plateaus indicates sequencing depth was sufficient to capture most of that sample's diversity; one still rising steeply at the sampling depth used suggests deeper sequencing would likely reveal more.



Sample classifier results

Tests whether community composition alone can predict a sample's metadata category, using a random-forest classifier. "Baseline accuracy" is what a naive classifier predicting only the most common class would achieve; an "accuracy ratio" above 1 means community composition carries real predictive signal for that column.

metadata column	metric	value
collection-date	Overall accuracy	0.455
collection-date	Baseline accuracy	0.273
collection-date	Accuracy ratio	1.667
country	Overall accuracy	0.909
country	Baseline accuracy	0.909
country	Accuracy ratio	1.000
env-medium	Overall accuracy	0.909
env-medium	Baseline accuracy	0.455
env-medium	Accuracy ratio	2.000
replicate	Overall accuracy	0.091
replicate	Baseline accuracy	0.545
replicate	Accuracy ratio	0.167